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ORIGINAL PAPER

Phleum pratense-pollen adaptive variations and pollen microbiome investigation under different climatic regions and prospects of allergenicity

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Abstract

Phleum pratense is an allergenic grass that pollinates in spring in Pakistan. Databases *Allergenonline.org* and *Allergen.org* record ten *P. pratense* allergens and their isoforms. Phl P 1, Phlp 5, and Phl p 11 are major *P. pratense*-pollen allergens with demonstrated basophil activity and skin test reactivity. Little is known about *P. pratense* pollen adaptive variations in different climatic regions and pollen-associated microbial diversity. In this study, we collected *P. pratense*-pollen and soils in the spring season 2022. Samples were collected from three climatic regions in Pakistan (R1, R2 and R3) with differences in mean monthly air temperature, mean monthly precipitation and elevation. The morphology of pollen was observed by light microscopy, scanning electron microscopy (SEM), biochemical fingerprint analysis, and composition of pollen were investigated by fourier-transform infrared spectroscopy (FTIR). The pollen-associated bacterial populations were identified through a Biolog GEN III microplate system. The pollen water-soluble proteins were isolated and stabilized in phosphate buffer saline (PBS) and tested for allergenicity responses through dot blots and western blots analysis. The morphology study found difference in pollen biochemical composition. Biolog identified *Brevibacterium epidermidis* and *Pantoea agglomerans* from *P. pratense* pollen. Protein extract quantification and sodium dodecyl sulfate-poly acrylamide gel electrophoresis (SDS-PAGE) gel found decreased protein expression in R1 region pollen compared to R2 and R3 region pollen. Allergenicity studies found differential expression of beta-expansin and profilin allergens in pollen obtained from the three regions. Beta-expansin and profilin were suppressed in R1 pollen and expressed in compared to R2 and R3 pollen. This is the first study to identify *B. epidermidis* and *P. agglomerans* growth on *P. pratense* pollen. Variable allergen expression in *P. pratense* pollen has also been observed in different regions. Soil pH, an increase in mean monthly temperature and a decrease in mean monthly precipitation correlated with pollen biochemical composition, and reduced beta-expansin and profilin expression involved in pollen growth and development. The findings of this research are unique, which enhances basic knowledge and understanding of *P. pratense*-pollen associated microbiota and climate change impacts on the pollen allergen expression.

Keywords Climate change · *P. pratense* · Biolog · *Brevibacterium epidermidis* · Allergen · Pollen

Introduction

Phleum pratense (Timothy grass), a member of the Poaceae family, which causes allergies in susceptible populations in many countries (Sekerkova et al. 2012; Mendy and Zeldin 2020; Beggs 2021; Xu et al. 2021) and accounts for 15% and 20% of grass pollen sensitization in the United States of America and Europe (García-Mozo 2017). A single grass pollen contains 0.040 ng allergen that activates human immune system (Jung et al. 2021). Water-soluble proteins of the pollen encounter the mucosal layer of the nasal cavity

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during inhalation, and induce the immune system, resulting in sneezing and release of watery fluids from the nasal cavity. So far, 10 allergens (Phl p 1, Phl p 2, Phl p 3, Phl p 4, Phl p 5, Phl p 6, Phl p 7, Phl p 11, Phl p 12, and Phl p 13), and their isoforms have been reported from grass pollen (*Allergen.org)*, (García-Mozo 2017). Out of these 10 various pollen allergens, Phl p 1, Phl p 2 and Phl p 5, cause 50–95% *P. pratense* pollen allergies (Hoffmann et al. 2022). Individual pollen allergens of grass are very well characterised and specific bacterial communities from the pollen have been reported (Obersteiner et al. 2016). The birch tree and *P. pratense*-microbial diversity correlation to allergenicity factors have been identified (Bet v 1/ Phl P 5 pollen associated lipid mediators). Inhalation of bacterial endotoxins (lipopolysaccharide) has been reported in lung inflammation (Cardinale et al. 2019). Similar to lipopolysaccharide gram-positive bacteria supernatants may induce allergic immune responses in human beings. Dysbiosis of the ocular surface microbiome plays an important role in ocular pathogenesis (Cavuoto and Zhu 2022). Epithelial layer dysfunction role has been indicated in rhinitis, atopic dermatitis, food allergies and asthma (Singh et al. 2022) coeliac disease, eosinophilic esophagitis and inflammatory bowel disease (Akdis 2021). *P. pratense* pollen may contain microbiota that could be involved in pollen allergy.

P. pratense grows in different climatic regions of the globe and the allergen expression of *P. pratense* under different climatic conditions is important to investigate climate change effect on the grass pollenallergen expression. Climate change is one of the biggest global challenges (Beggs 2021), and studies have indicated that it affects pollen chemical composition (Zimmermann et al. 2017), pollination season and pollen allergies (Subiza et al. 2019; Sapkota et al. 2019, 2020; Beggs 2021). Water stress in cereals causes pollen infertility. Soil pH and air quality affect pollen size (Lau and Stephenson 1993; Verscheure et al. 2023). Soil pH conditions affect *Ambrosia artemisifolia L.* growth and pollen IgE binding signal strength (Gentili et al. 2018). An increase in temperature (~5 °C) above the optimum growth conditions of a plant inhibits mRNAs and proteins synthesis (Shyam S. Mohapatra 1996). Temperature increase causes immature bursting of the tapetum layer of the pollen due to oxidative stress that makes pollen infertile. Oxidative stress in ryegrass a member of Poaceae due to variant pollution levels affects pollen allergenicity (Lucas et al. 2019). Ryegrass responds differently to drought and elevated air temperature. *P. pratense* produces less pollen allergen content in response to drought and elevated air temperature (Jung et al. 2021). Pollen biochemical composition (proteins, lipids and carbohydrates) adaptation varies in response to different environmental conditions, especially temperature (Zimmermann et al. 2017). Climate change impact on aeroallergen growth and development, concentration and distribution, and associated allergies depict increase in asthma and allergy patients hospitalizations (Beggs 2021). The impact climate change has on *P. pratense*-pollen allergen expression and their biochemical composition and sensitization has not been studied so far. The grass grows in different climatic regions of Pakistan, which is reckoned among the top most affected countries due to climate change. This study explores adaptive variations in *P. pratense* pollen and associated allergies, and *P. pratense* pollen-associated microbiome in three different climatic regions of Pakistan. We profile the effect of mean monthly temperature, mean monthly precipitation, soil pH conditions and elevation from sea level may affect pollen morphological features and allergen expression.

Materials and methods

Pollen sampling

P. pratense pollen and earth surface soils of respective locations were sampled in sterile zip lock bags in March 2022 from three distinct geographic regions of Pakistan. 5 g of the soil sample was suspended in 10 ml of deionized water from each region to determine the soil pH through Dynamica pHMaster pH meter serial #14,221,009 (McLean 1982). Regional altitudes were measured through a digital altimeter.

Region 1 (district Peshawar), region 2 (district Islamabad) and region 3 (district Kotli) are located at 331 m, 507 m and 609 m elevation from sea level, respectively. According to the Koppen-Geiger Climate Classification, Region 1 has BSh (dry, semi-arid and hot), whilst Regions 2 and 3 have Cwa (temperate, dry winter and hot summer) weather conditions (Adnan et al. 2017) (S1. Fig, https://ndmc.pmd.gov.pk/new/annual-climate.php?p=climate-classification). The mean monthly temperature during March 2022 for Region 1, Region 2 and Region 3 was 22.9 °C, 21.9 °C and 22.1 °C, respectively, and mean monthly precipitation during the period was 5.2 mm for R1, 59.5 mm for R2, and 21.1 mm for R3 (https://cdpc.pmd.gov.pk/Pakistan_Monthly_Climate_Summary_March_2022.pdf). Mean annual temperature and precipitation data from the past decade for the three regions was retrieved from Pakistan Meteorological Department and analysed for the past decade climatic trend. Two-way-ANOVA was applied on the mean annual temperature and precipitation dataset (2010–2021) of R1, R2 and R3 (S1. Table), and Tukey's multiple comparison test was applied to the means of pairs. For each region, mature pollen were collected from 3 to 5 nearby-located *P. pratense* plants on a sunny day by clipping the inflorescence of the

plant in sterile 50 ml falcon tubes. Only mature flowers of the plant were sampled, on a bright sunny day, to get fully developed pollen (about to spread in air). Samples were air dried at room temperature at a relative humidity of 30 g. m^{-3} for one week, and then tipped on a 500-µm pore size mesh. Isolated pollen from the inflorescence were passed on a 100-µm pore size mesh to remove non-pollen floral parts and optimize pure pollen concentration. Pure pollen samples were processed immediately or stored at room temperature in 1.5 ml sterile tubes in a wooden dark cabinet until further use.

Microscopy

Pollen grains (10 mg) were soaked in iso-propyl-alcohol (200 µl) and aqueous ponseau S stain for ten minutes to rehydrate. Folding effect was reduced by addition of aqueous ponseau S stain that rehydrated dry pollen and stained pollen red for microscopy. To minimize the folding effect, 30 pollen grains diameter readings were recorded for every region and the mean value for every region are presented respectively. Then 10 µl of the mixture was spread on a microscopic glass slide and was air-dried. The glass slide was fixed with 100% pre-heated glycerine and was visualised under a Zeiss light microscope at 20 X and 100 X (Whitney and Needham 2014). Pollen diameter was calculated through Image-J software (153-win-java8). Scanning electron microscopy (SEM) of dried *P. pratense* pollen was done on a HITACHI SU 1500 equipment. Pollen was directly kept on carbon patch-covered SEM stub holders. For each region 20 pollen diameters were calculated through SEM microscale tool at 50 µm resolution at 1 KV accelerating voltage in a high vacuum mode (Depciuch et al. 2016). Two-way ANOVA was used to test variance in pollen diameter of the three regions, and Tukey's multiple comparison test was applied to the means of pairs (S2 Fig. and S1. Table).

Pollen fourier transform infrared spectroscopy

Dried pollen grains (20 mg) of R1, R2 and R3 were placed on fourier transform infrared spectroscopy (FTIR) IRTracer-100 knob for FTIR analysis (Kenđel and Zimmermann 2020). Vertex 70 V FTIR spectrometer measurements were taken using the attenuated total reflectance (ATR) method. Resolution of 4 cm^{-1} was selected for infrared radiations in the range of 500–4500 cm^{-1} to draw peaks (Depciuch et al. 2018). Pollen samples were collected from the three regions, and each region sample was scanned three times. Normalized obtained peaks and saved data files in comma-separated values (CSV) format that were processed for drawing FTIR peaks through Origin pro 8.5 E-2018 software for spectral peaks and principal component analysis (PCA) (Kenđel 2020), and one-way-ANOVA. Spectral datasets regions for protein (1500–1700 cm^{-1}) and lipid (2820–3000 cm^{-1}) were magnified from the entire FTIR spectra to make R1, R2 and R3 comparisons. The entire FTIR spectral region (500–4000 cm^{-1}) datasets were used for PCA and one-way ANOVA.

Identification of pollen associated bacteria

P. pratense pollen samples from the selected regions were shaken for 30 min in 1 ml of 0.9% saline four-fold dilutions were made. 100 µl of the last dilution was plated on Luria Bertani (LB) agar plate in an ESCO class II biosafety cabinet (SC2-4E1). Plates were incubated for 48 h at 33 °C. Morphologically similar colonies appeared on plates from all three samples of R1, R2 and R3 regions within 24 h. The bacteria were purified and glycerol stocks were prepared. The purified bacterial culture was subjected to identification through Biolog GEN III system (Biolog, Hayward, CA, USA) to test the bacteria's ability to utilize 71 carbon sources present in the microplate. Sub-cultured single colony from 24-hour bacterial culture on Biolog Universal Growth (BUG) media. A single colony from the BUG media petri plate through a sterile inoculator was taken and the inoculum swirled in inoculation fluid A (IF-A) to release the bacteria colony in the IF-A. IF-A turbidity was checked before and after the release of bacterial colony in the fluid, to ensure that the transmittance value of the IF-A was above 90%. After that, 100 µl IF-A was transferred to each well of the GEN III microplate through a multichannel pipette. Plates were incubated at 33 °C for 24 h, and plate readings recorded visually from the colour readings after 6 h and 22 h of incubation. Colour change in the microplate wells was compared to the negative and positive controls, well #1 and well #10 respectively. The plate 22 hours' results were used for the identification of bacteria from Biolog database.

Protein extraction

Total soluble proteins of *P. pratense* dry pollen were extracted through defatting and non-defatting methods in 1X-phosphate buffer saline (PBS) (Cases et al. 2014) with some modifications. For defatting, suspended 1 g of pollen in 20 volumes of N-Hexane for each sample and kept the samples shaking at room temperature for 2 h, and filtered through grade 1 Whatman® filter paper 45 micrometre diameter. Defatted pollen was extracted in 1X-PBS, centrifuged and filtered through 0.45 µm pore size syringe filters. Aliquots of the filtrate were labelled, quantified and stored at -20 °C. The same process for non-defatting was repeated.

Two-way ANOVA was used to check variance in protein quantities.

Human sera selection

The allergenic human sera to *P. pratense* listed in Table 1 was purchased from Plasma Lab International (3128 Norton Avenue, Everett, WA 98,201 USA http://www.plasmalab.com. Also, *P. pratense* allergenic sera available in FARRP Lab University Of Nebraska Lincoln USA were used, and serum was collected from Pakistani students at the University of Nebraska Lincoln USA (Ramadan et al. 2021). Ethical consents from donors were taken to use their serum in research following approval by the collecting source of serum donors as required by the Institutional Review Board (IRB) of the University of Nebraska-Lincoln.

Allergenicity testing

The allergenicity of the pollen protein extract was confirmed by IgE binding tests with selected sera through dot blot followed by western blot analysis (Ramadan et al. 2021) with slight modifications. Monoclonal mouse anti-human IgE conjugated with horseradish peroxidase (HRP) was used (Southern Biotech, Birmingham; AL clone B3102E8 Cat #9160–05) 1:1000 diluted in PBS-T. IgE binding was detected with Super-signal West Dura Extended Duration Chemi luminescent (ECL) substrate (Pierce, Rockford, IL, USA, Cat # 34,076) on a UVP Bio-Spectrum 815 Imaging System for 35 min. Manually set up image intensities to visualize and reduce the city background noise.

Table 1 Summary of the selected serum samples clinical history

Serum ID	Gender /Race	Reported allergies	Symptoms	ICap Total IgE	Specific IgE CAP		
					P. pratense	Birch	Ragweed
PL22241	M /Caucasian	Cat dander, dog dander, *P. pratense* pollen ragweed pollen	Allergic rhinitis and FEIA	453	7	0.6	17.4
PL22272	M /Caucasian	Ragweed, birch pollen	Allergic rhinitis and FEIA	NA	1.1	2.6	11.2
PL22887	M /Caucasian	Cat, Soy, hazelnut, almond, walnut, *P. pratense* pollen, ragweed, sunflower	Allergic rhinitis, FEIA	354	13	5.8	7.8
PL 24,180-D	F /American	Dog and cat dander, *P. pratense*	Allergic rhinitis, asthma	135	10.5	1.53	0
PL 24,329-I	F /Caucasian	Multiple allergies	Allergic rhinitis	378	31.8	0	0
PL 25,218-G	M /American	Grass pollen	Allergic rhinitis	413	48.4	0	0
PL 29,148	M /Caucasian	NA	Allergic rhinitis and asthma and FEIA	525	0	0	0
19,362-CM	F /Caucasian	*A. hypogea*, *orchard grass*, *P. pratense*, *birch*, *ragweed*, *mugwort*	NA	377	61.6	20	31.2
23,508-JK	M /Caucasian	NA	NA	300	10.4	3.42	0
PL 27,121	M /Caucasian	NA	NA	4054	0	0	61
22,329 JE	M /American	NA	NA	*1953*	100	13.1	34
19,392 CS	NA/NA	*Soy*, *A. hypogea*		NA	0	0	0
9735 RE	NA/NA	NA	Anaphylaxis to peanut.	NA	0	0	0
22,104 JC	M /Caucasian	NA	Ragweed, house dust mite,	NA	0	0	0
PL27097	M /Caucasian	NA	*A. hypogea*, Birch, Ragweed	2692	0	87.7	18.1
PakPoln	M/ Caucasian	*P. pratense*, *B. papyrifera*, *C. sativa*	NA	NA	0	0	0

NA = Unknown information

Results

Analysis of mean annual temperature and precipitation data from years 2010–2021 by two-way-ANOVA at P value < 0.0001 showed significant differences in the three regions. Mean annual temperature was significantly different among the three regions. However, mean annual precipitation was significantly different between R1 and R2 and R1 and R3 regions. There was no significant difference between R2 and R3 (Fig. 1).

Pollen microscopy

SEM and light microscopy at 100 X found round shape of *P. pratense* pollen. Pollen exine layer has a scar like structure on each pollen that attached it to the pollen tube. The mean pollen diameter was 36.3 µm, 39.6 and 38 µm for R1, R2 and R3. Two-way ANOVA showed that R1, R2 and R3 pollen diameter means were significantly different at $P = 0.05$ (S2. Fig). R1, R2 and R3 soil pH was 7.3, 6.4 and 6.9 respectively.

Pollen fourier transform infrared spectroscopy (FTIR)

The pollen FTIR analysis of R1, R2 and R3 regions was done using the entire FTIR spectra (wavelength of 500–4000 cm^{-1}). In the entire FTIR spectra, proteins region (spectral wave length 1700–1500 cm^{-1}) and lipids region (spectral wave length 3000–2820 cm^{-1}) showed variation in proteins and lipids functional group peaks (Fig. 2I and II). The range of peak transmittance values for proteins were 12–19 (Fig. 2-I) and for lipids 10–25 (Fig. 2 II) in the respective spectra which differed for pollen from all three regions. The transmittance values were high for R1 region having elevated mean annual temperature and low mean annual precipitation in comparison to R2 and R3. Proteins and lipids functional groups spectral zones along with peak frequencies and their chemical bonds (Depciuch et al. 2017), are given in Table 2. FTIR PCA of the three regions datasets showed differences in the three datasets. PCA score of the three datasets is more than 0.3 on the score plot which marks the difference in the three datasets (Fig. 2 III). The PCA value of R1 FTIR differs from R2 and R3 by 0.6 score which marks the dataset is different from R2 and R3 FTIR datasets. PCA analysis of entire FTIR datasets for

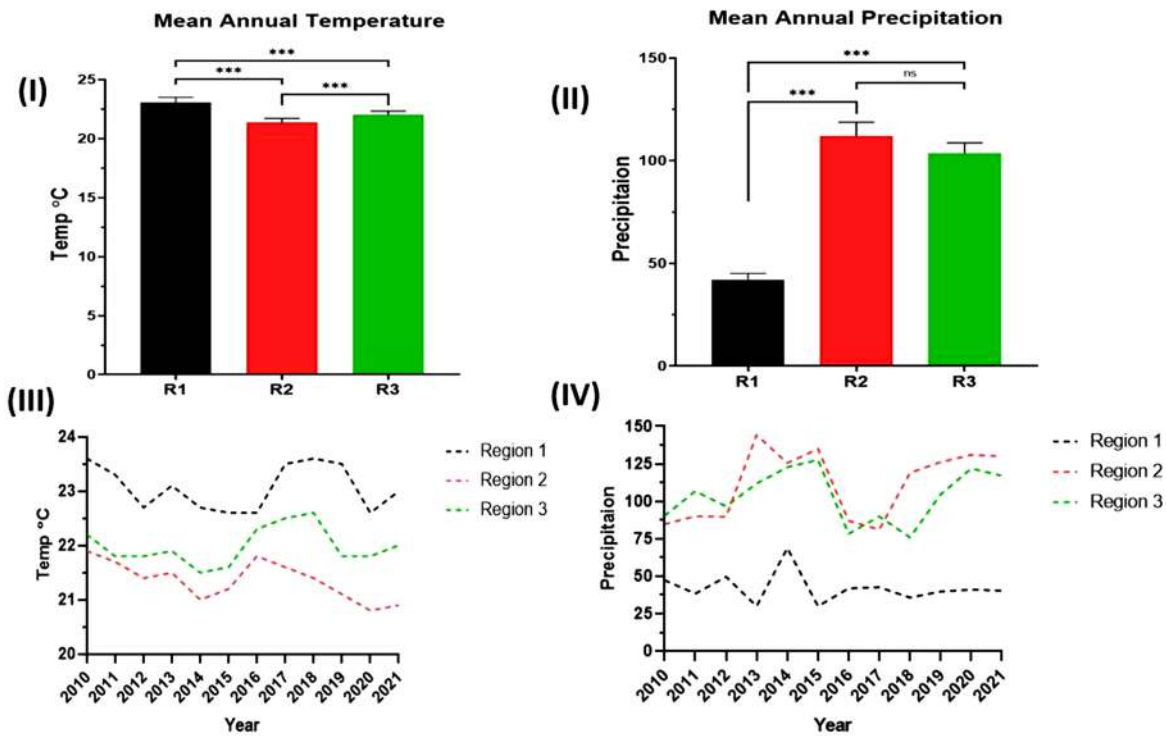

Fig. 1 Shows two-way-ANOVA and line graph of regions R1, R2, and R3 mean annual temperature (°C), and mean annual precipitation (mm) from year 2010 to 2021. (I) display two-way-ANOVA for mean annual temperature. Y-axis shows temperature, X-axis shows regions, and the star symbols on the top of the image show level of significance. (II) display two-way-ANOVA for mean annual precipitation. Y-axis shows precipitation, X-axis shows regions and the star symbols on the top of the image show level of significance. (III) shows trend of mean annual temperature and IV) shows trend of mean annual precipitation

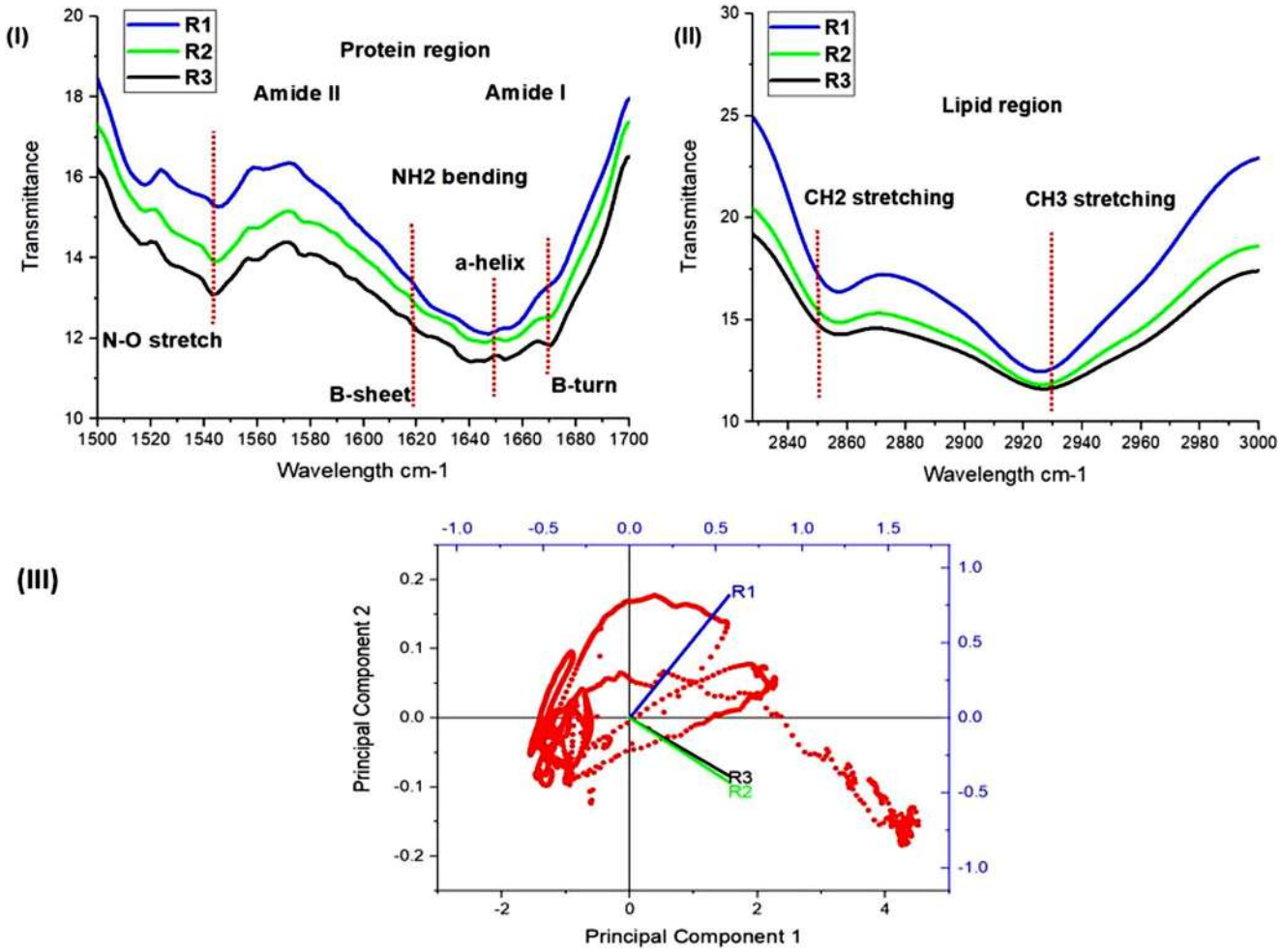

Fig. 2 FTIR analysis of *P. pratense* pollen collected from three regions R1, R2 and R3.(I) Shows *P. pratense* pollen FTIR spectral peak of protein region. (II) shows spectral peak of lipid region. In Fig. I and II, the Y-axis denotes transmittance values, and the X-axis shows wavelength. Blue, green and black colour show pollen samples of R1, R2 and R3 regions respectively. The dotted line shows spectral peak differences at a specific functional group within the protein region. (III) shows PCA of the FTIR data sets. The red dots show data points while the three lines R1, R2 and R3 show vector forms of the datasets. Principal component I and principal component 2 are transmittance values of R1, R2 and R3 regions FTIR datasets shown in vector form. The value from 0.0–1.0 shows PCA score

Table 2 Spectral zone, functional group frequencies, and associated chemical bonds

Spectral zone	Peak frequency cm^{-1}	Chemical bonds
Amide I 1700–1600 cm^{-1}	1620 β-sheet 1650 α-helix	NH_2 bending
Amide II 1600–1500 cm^{-1}	1670 β-turn 1550	N-O stretch
Lipids	2930 2850	CH_3 stretching (lipids) CH_2 stretching (lipids)

all functional groups' regions (500–4000 cm^{-1}) is shown in Fig. 2 III which marked differences in the three datasets. Pollen FTIR one-way-ANOVA of entire FTIR datasets for all functional groups' regions (500–4000 cm^{-1}) showed that the population means are significantly different at $P \geq 0.05$ (S3. Fig).

Pollen associated bacteria Biolog identification

Biolog identification system using GEN III microplate identified *B. epidermidis and P. agglomerans*. The plate has 71 carbon sources, negative and positive controls. The bacteria utilized completely 1% NaCl, 4% NaCl, 8% NaCl, 1% sodium lactate, guanidine HCl, nalidixic acid, lithium chloride, aztreonam, sodium butyrate, and sodium bromate. Complete utilization of the mentioned media developed colour similar to well A10 (positive control). The bacteria completely or partially failed to utilize D-raffinose, D-sorbitol, pectin, α-D-lactose, D-maltose, D-melibiose, L-galactonic acid lactone, D-trehalose, β-methyl-D-glucoside, D-cellobiose, D-salicin, sucrose, N-acetyl-β-D-mannosamine, D-turanose, N-acetyl-β-D-galactosamine, D-aspartic acid, stachyose, D-serine,

troleandomycin, lincomycin, vancomycin, fusidic acid, rifamysin SV, tetrazolium violet, D-serine, minocycline, niaproof 4 and tetrazolium blue. Complete failure to utilize the media gave no colour similar to well A1 (negative control). The bacteria utilized the remaining 71 carbon sources partially and developed light purple colour. The dark purple-coloured wells showed strong catalase positive reaction, light coloured wells depicted slight catalase positive, and no colour development showed oxidase negative characters of the plate. Comparison of the nutrient utilization and the colour development pattern identified the bacteria as *B. epidermidis and P. agglomerans.*

Pollen protein extraction

Total pollen protein extracted in 1X PBS through defatting and non-defatting shows no significant difference in the two extraction methods. Protein quantities detected through non-defatting and defatting methods in 20 µl volume were (2.3 µg and 3.6 µg), (8.0 µg and 8.3 µg) and (6.6 µg and 7.1 µg) for R1, R2 and R3 respectively. R1 has the lowest protein quantity, while R2 has the highest protein quantity. Two-way ANOVA demonstrated that pollen protein extracts from the three regions were significantly different ($P = 0.05$).

Dot blot analysis

Pollen protein extracts serum IgE binding results through dot blot showed allergenicity towards 16 human sera (see Fig. 3). Sera having strong IgE binding showed intense reaction and large diameter dots at 3 min exposure under UVP imager for pure protein extracts and their 1:10 dilution in 1X PBS. Sera having weak IgE binding showed faint reaction and small dot diameter under UVP imager. Less IgE reactive sera showed IgE binding in dot blots only for pure protein and did not show IgE binding with diluted protein extracts. Strong IgE reactive sera showed IgE binding to 1:10 protein dilutions. Pure protein extracts and diluted protein extracts IgE binding is given in Fig. 3.

SDS-PAGE pollen proteins profile

R1, R2 and R3 pollen protein extracts resolved on Invitrogen 10–20% tris glycine premade gel showed differential banding pattern at non-denaturing and denaturing conditions. R1 having less protein concentrations showed fewer and faint bands at non-denaturing and denaturing conditions. R2 and R3 showed similar banding pattern at non-denaturing and denaturing conditions.

Western blot analysis

Pollen protein extracts in 1X PBS were loaded (4 µg /15µl) to each well, SDS PAGE was run, and proteins were transferred to NC membrane at a constant voltage of 30 volts and run for 90 min. Primary antibody, (300 µl serum samples) were diluted into (1:10). The secondary antibody (200 µl), HRP mouse conjugated antihuman IgE Southern Biotech was diluted into (1:1000). NC membrane showed allergen

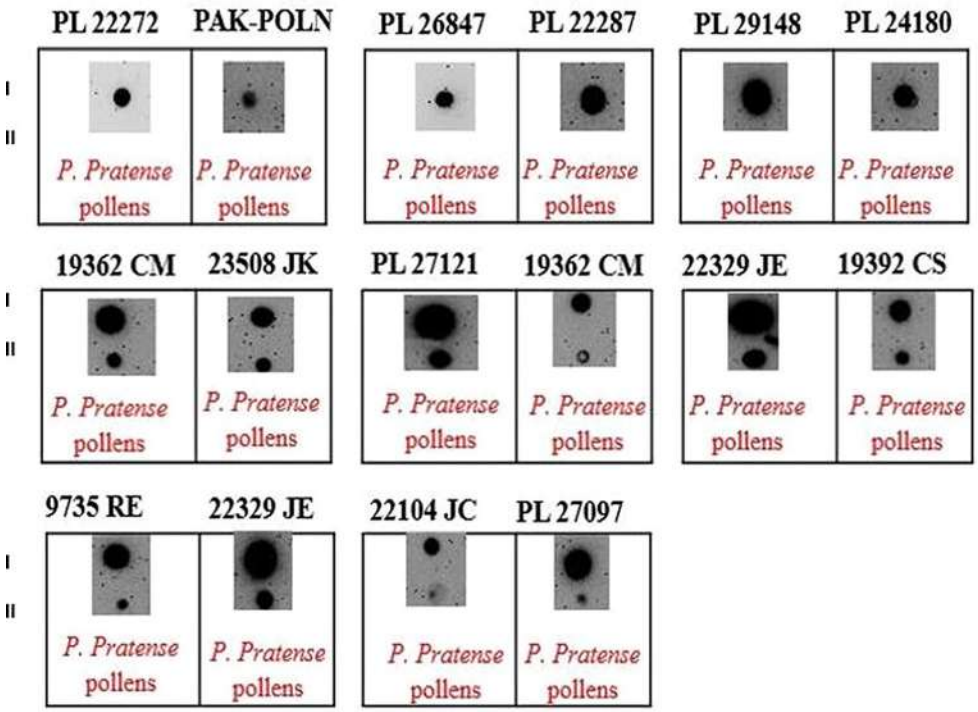

Fig. 3 *P. pratense* pollen-protein dot blot analysis. The dot in the box shows pollen allergen reactivity with primary antibody present in the serum of study participants. The bigger dots show pure pollen protein IgE binding, while the smaller dot denotes 1:10 1X PBS diluted protein IgE binding. The number above the box represents the serum identification number used in the study. Pure and 1:10 diluted pollen proteins IgE binding is shown by roman numbers I and II respectively on the left side of the image

bands under UVP imager. PL 22,287 serum showed about 32 kDa bands putative Phl p 1 allergen for R2 and R3 (Fig. 4I). For Pak Poln serum and 26,847, proteins transferred to NC membrane showed putative allergen, either Phl p 2 or Phl p 3 for R2 and R3 (Fig. 4II, and Fig. 4III). For 9735 RE serum, proteins transferred to NC membrane showed putative allergen, either Phl p 2 or Phl p 3 for R2 and R3. R 3 showed two extra bands of about 14 kDa and 32 kDa. These bands show putative Phl P 1 or Phl p 5 (Fig. 4IV).

Proteins transferred to NC membrane showed bands for R2 and R3. 27,121 RE serum (300 µl) used as primary antibody, and diluted 1:10. The secondary antibody, HRP mouse conjugated antihuman IgE Southern Biotech 1:1000 diluted showed about 10 kDa bands for R2 and R3, and R2 showed additional allergens about 12 kDa and 14 kDa at 3 min exposure under UVP imager. The same NC membrane showed about 55 kDa allergen Phl p 4 for R2 and R3 samples (Fig. 5).

Discussion

Pollen proteins can easily come across the nasal epithelial cavity during breathing, where pollen allergens cause allergic reactions. Pollen based allergies are increasing with climate change due to pollen season extension, increases in the amount of pollen and changes in allergenic protein concentrations (Ziska 2021). Climatic conditions have also a pronounced effect on pollen morphological and biochemical features, and protein expression profiling. Air quality, soil and water conditions affect pollen size (Lau and Stephenson

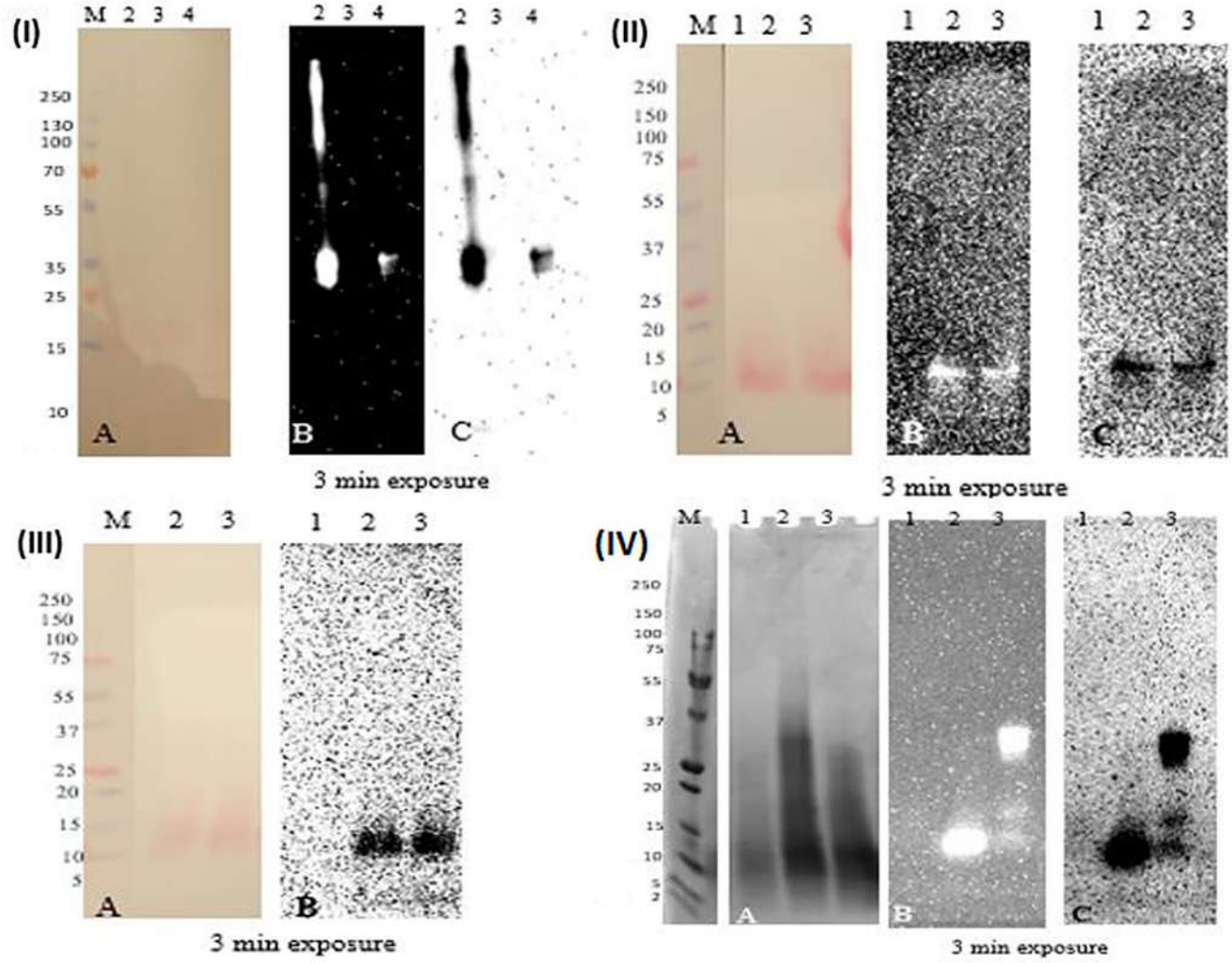

Fig. 4 *P. pratense*-pollen western blotting results for serum samples PL 22,287, Pak Poln, 26,847 and 9735 RE respectively (Table 1). In the fig I, II, III and IV, A shows protein transfer to NC, B shows allergens on NC visualised under UVP Imager at 3 min exposure while C shows inverted image of B. M is precision plus protein marker. Lanes 1, 2 and 3 are R1, R2 and R3 pollen proteins loaded at 25 °C non-reducing conditions

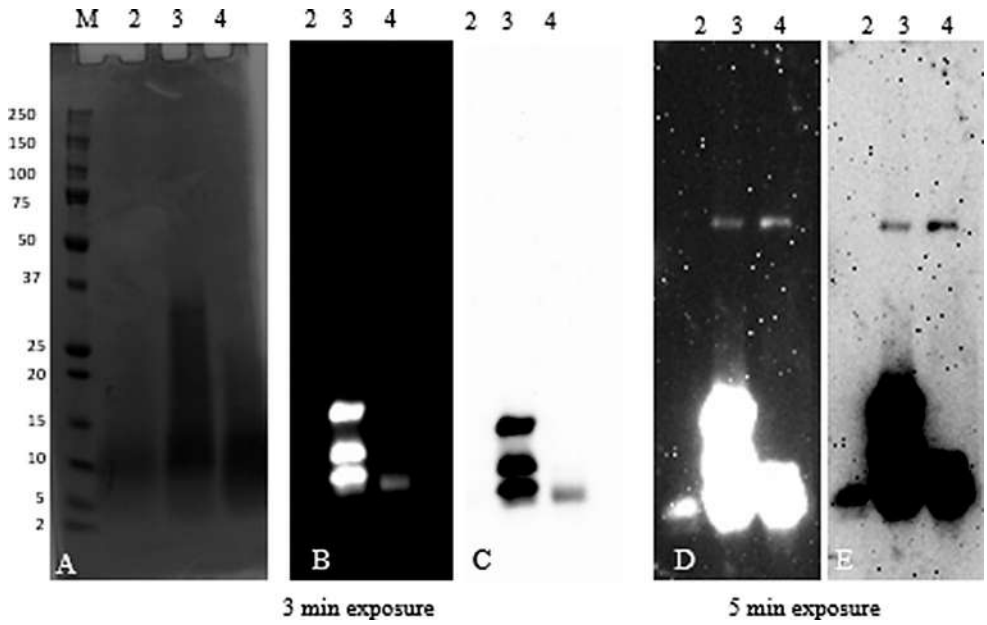

Fig. 5 *P. pratense*-pollen western blotting results for serum 27,121 (Table 1). Pollen protein extracts in 1X PBS were loaded (4 µg /15µl) to each well, SDS PAGE was run, and proteins were transferred to nitrocellulose membrane at a constant voltage of 30 volts and run for 90 min. A shows protein transfer to NC, B shows allergens on NC visualised under UVP Imager at 3 min exposure while C shows inverted image of B. D shows the same western blot visualized under 5 min exposure while E is its inverted image of D. M is precision plus protein marker. Lanes-2, 3 and 4 are R1, R2 and R3 pollen proteins loaded at 25 °C non-reducing conditions

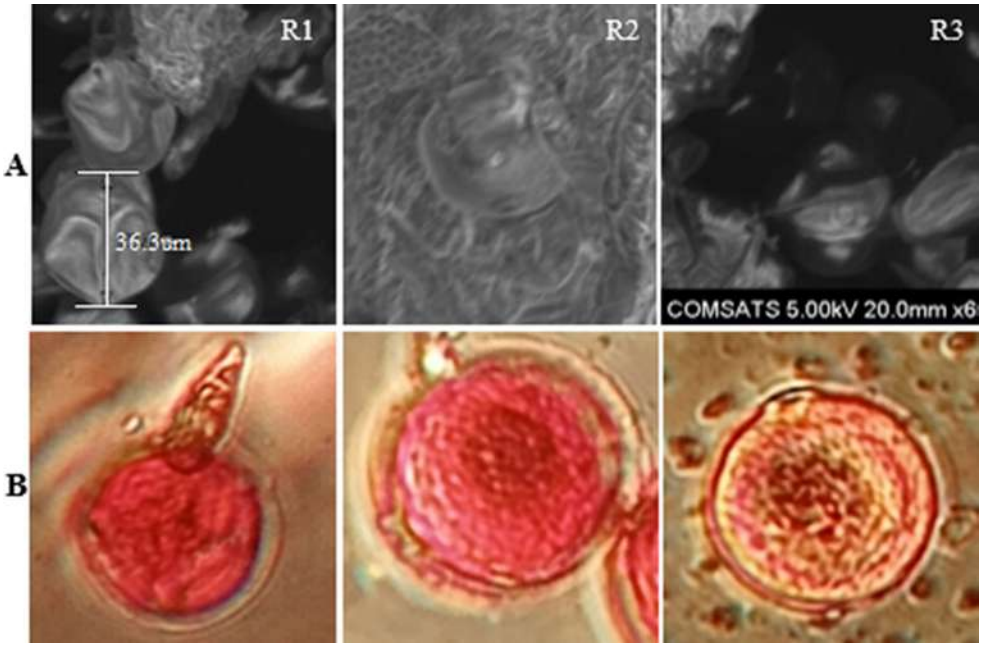

Fig. 6 Light microscope analysis of *P. pratense* pollen at 100X resolution from three climatic regions R1, R2 and R3 (**A**) Scanning electron microscopy (SEM) analysis, the black line in the centre shows the average diameter of pollen calculated through SEM in-built microscale. (**B**) light microscopy images of pollen of R1, R2 and R3 regions coloured with ponseau S stain and captured through light microscope having circular and dotted morphology

1993; Verscheure et al. 2023). Environmental conditions have been found variable on different species pollen. *P. pratense* is an abundant perennial grass that produces allergenic pollen. In this study, we have found that significant differences are present in the environmental conditions of the three regions (see Fig. 1). These regions elevation from sea level, soil pH, mean annual temperature and mean annual precipitation vary from each other. Our study findings are synonymous with the findings of Adnan et al. (2017), who established that these three regions occur in three distinct agro-climatic zones. They have divided the whole region of Pakistan into 8 distinct agro-climatic zones based on observational data of precipitation and evapotranspiration from 1951 to 2010. We have investigated the effect of mean monthly temperature, mean monthly precipitation, elevation from sea level, and soil pH under natural conditions in three different regions (R1, R2 and R3) on *P. pratense* pollen biochemical composition, and pollen allergen expression. The mean pollen diameter of the R1 region (36.3 µm) was smaller than R2 (39.6 µm) and R3 (38 µm) (see Fig. 6). Pollen diameters were significantly different in all regions ($P = 0.05$). R1 region has increased mean monthly temperature and decreased mean monthly precipitation in comparison to R2 and R3 regions. In previous investigations (Jung et al. 2021), it was observed that decreases in mean pollen diameter (lower size & weight) of *P. pratense* was in

response to drought stress. There were no other significant differences in pollen grain structural morphology, and the microscopic identification of pollen cannot differentiate pollen to a species (Radaeski et al. 2016). Soil pH of R1, R2 and R3 was 7.3, 6.4 and 6.9 respectively. The soil pH may have affected pollen allergen expression. In a previous study conducted on *Ambrosia artemisifolia L.*, soil pH growth was associated with pollen IgE binding signal. The plant pollen protein developed at pH 5 soil has higher IgE binding signal in comparison to pollen developed at pH6 and pH7 (Gentili et al. 2018).

Water soluble protein content was lower in R1 region pollen in comparison to R2 and R3. These findings are similar to those of Jung et al. (2021), where they showed that *P. pratense* pollen has lower protein and allergen content under drought stress conditions. It was observed that 0.87 °C warming treatment according to representative concentration pathway (RCP) 2.6, yielded 2.4 ng protein and 0.029 ng allergen/per grain in comparison to the control group 3.1 ng protein and 0.040 ng allergen/per grain (Jung et al. 2021). Under drought stress conditions, *P. pratense* pollen showed lower size and weight under drought stress condition. Our results showed that R1 (dry semi-arid conditions, S1. Fig) pollen protein concentration was 3.6 µg/10 µl, in comparison to R2 and R3 (7.1 µg/10 µl and 8.3 µg/10 µl) respectively. Overall, differences in pollen protein concentration, associate to variation in environmental conditions of different regions.

FTIR analysis showed differences in pollen biochemical profile as the spectral peaks differ in pollen of all three regions (see Fig. 2-I-II). Region R1 has elevated mean monthly temperature and low mean monthly precipitation (22.9 °C, 5.2 mm) but lower elevation from sea level in comparison to R2 (21.9 °C, 59.5 mm) and R3 (22.1 °C, 21.1 mm). Region R1 has elevated mean monthly temperature and low mean monthly precipitation (22.9 °C, 5.2 mm) but lower elevation from sea level in comparison to R2 (21.9 °C, 59.5 mm) and R3 (22.1 °C, 21.1 mm). Therefore, FTIR spectral peaks reflected that R1 pollen transmittance values were high in comparison to R2 and R3 regions. R2 and R3 regions have almost similar temperature and precipitation conditions but differ in elevation. However, the pollen of these two regions showed similar spectral peaks. Spectral differences in pollen growing in different environmental conditions correlate to differences in pollen protein, carbohydrate, lipids and sporopollenin (Zimmermann and Kohler 2014). They studied 300 different plant species pollen collected from variant environmental regions and found differences in pollen FTIR peaks within single species pollen collected from different environmental regions. Our results indicated that spectral bands differ at 1540 cm^{-1} and 1670 cm^{-1} respectively for R1, R2 and R3 (Fig. 2-I). Spectral bands at 1540 cm^{-1} and 1670 cm^{-1} correspond to amide-II and amide-I respectively (Diehn et al. 2020). Similarly, 2850 cm^{-1} and 2930 cm^{-1} spectral bands correspond to lipids, and the spectral bands differ at these wavelengths for R1, R2 and R3. Similarly, one-way ANOVA for R1, R2 and R3 FTIR datasets of entire spectral regions showed that population means are significantly different at $P \geq 0.05$ (S3. Fig). These differences in pollen FTIR peaks associated with different regions' environmental conditions. Environment interactions influence pollen biochemical composition and pollen phenotypic plasticity occurred due to locational environment (Bagcoglu et al. 2017) causing variations in pollen proteins, lipids, sporopollenins and carbohydrates. Moreover, pollen associated bacteria may influence pollen phenotypic plasticity.

We isolated gram positive, catalase positive and oxidase negative and gram negative, oxidase negative and catalase positive bacteria from *P. pratense* pollen, having optimal growth at 33 °C. The bacteria formed grey white and yellow colonies on LB agar for 24 h of incubation period. Biolog Gen III plate testing identified the bacteria as *B. epidermidis and P. agglomerans*. Previously, *B. epidermidis* was discovered from cheese, milk and human skin isolates that formed 2–4 mm grey white colonies on nutrient agar media after 24 h incubation between 30 and 37 °C. The bacterium was characterized as catalase positive, oxidase and urease negative, non-acid fast, rod shaped or coccoid and non-motile bacteria (Collins et al. 1983). Brevibacteria species are opportunistic pathogens mostly isolated from clinical samples (Wauters et al. 2004). *Staphylococcus epidermidis* and *B. epidermidis* are classified as risk group II and risk group I respectively that cause negative pulmonary health effects (Martin et al. 2010). *P. agglomerans*, an opportunistic pathogen was identified from the flowers of *Epipactis helleborine* that strengthens the findings of our study (Jakubska-Busse et al. 2021). Comparisons of allergenic and non-allergenic plants pollen and pollen associated bacterial endotoxin analysis showed that pollen related microbiota influence pollen allergic potential (Cardinale et al. 2019). Similar to our findings, only *Bacillus* species was identified from *P. pratense* pollen sampled from 20 different urban and rural sites (Obersteiner et al. 2016). These findings strengthen our hypothesis that pollen associated microbes may play an important role in pollen allergies. Allergenic pollen associated microbiome role should be studied further to explore pollen allergies. Furthermore, pollen allergen expression in respect to locational environment is also important for *P. pratense*-pollen allergies.

The pollen protein extracts from all regions resolved on 10% tris glycine gels found similar protein banding pattern for R2 and R3 regions. However, the R1 banding pattern was different and lower than R2 and R3 regions

pollen (see Fig. 7). R2 and R3 have almost similar temperature and precipitation conditions but different elevations. R1 has different temperature, precipitation and elevation in comparison to R2 and R3. In the presence of adequate humidity, an increase in temperature induces vegetative and generative functions in Poaceae (Makra et al. 2014). *P. pratense* is early flowering specie but nonavailability of pollen count data from all regions limits the study to analyse pollen production timings. Similarly, it has been found that Poaceae flowering phenology and pollen count is affected by rainfall, maximum temperature, wind speed and humidity (Norris-Hill 1997; Jato et al. 2009). Rainfall affects total pollen count as compared to temperature (Jato et al. 2009). Air temperature facilitates anther opening, pollen release and pollen dispersal (Ščevková et al. 2020). The difference in mean monthly temperature, mean monthly precipitation, elevation from sea level and soil pH conditions in the three regions have caused differences in the PBS soluble protein expression. Water stress, heat stress and air pollution induce infertility in pollen (Shyam S. Mohapatra 1996). R1 has about 1.5 °C and 1.0 °C higher temperature R2 and R3 respectively (Humayun et al. 2023 Research square preprint). Pollen tube growth and development is sensitive to temperature stress. Heat shock proteins (HSPs) are active in immature pollen in the absence of heat stress. Response of HSPs decrease in mature pollen due to heat stress (Mascarenhas and Crone 1996). The decrease in *P. pratense*-pollen weight, and protein content variation under drought and heat conditions has been reported by Jung et al. (2021). The authors found significantly increased allergen content in control treatments as compared to heat treatments (warming treatment, warming + drought treatment) (Jung et al. 2021). Our study finds that putative allergens' expression in *P. pratense* pollen has also been changed in response to change in climatic conditions in the three regions.

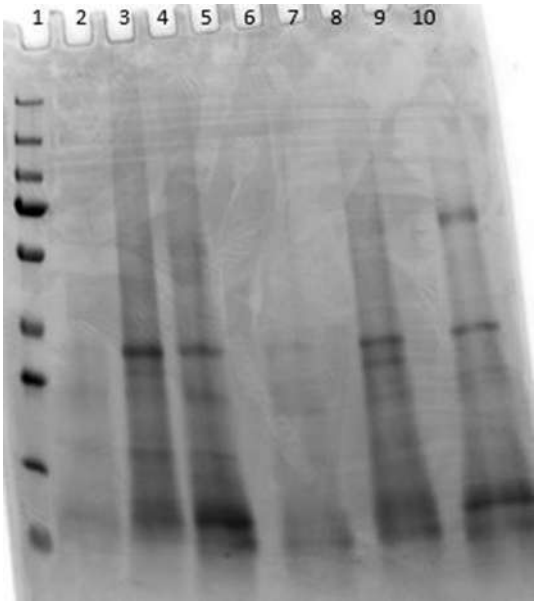

Fig. 7 Invitrogen 10–20% tris glycine premade gel run for 90 min of *P. pratense* pollen protein under 130-volt constant voltage. *P. pratense* defatted pollen protein (4 µg /15µl) extracted in 1X PBS were loaded in each well. Lane 1 is precision plus protein marker, lane 2 is *P. pratense* Peshawar 2022, lane 3 is *P. pratense* Islamabad 2022, lane 4 is *P. pratense* Kotli 2022 loaded at 25 °C under non-reducing conditions. Lanes 5, 7 and 9 are blank. Lane 6 is *P. pratense* Peshawar 2022, lane 8 is *P. pratense* Islamabad 2022, and lane 10 is *P. pratense* Kotli 2022, loaded at 95 °C and under reducing conditions

Decrease in protein quantity and intensity has caused low or no expression of putative allergens in R1 in comparison to R2 and R3 regions for several *P. pratense*-pollen allergens. Phl p 1 (32 kDa allergen) is a beta expansin involved in cell wall loosening during pollen tube growth (Sampedro and Cosgrove 2005). Phl p1 allergen expressed in R2 and R3, but its expression was missing in R1. Different environmental conditions of R1 affected Phl p 1 allergen compared to R2 and R3 (see Fig. 4I). Similarly, R2 and R3 have different mean monthly precipitation and elevation from R1.

Lol p 1 (27 kDa beta-expansin) is a major *Lolium perenne* allergen, and a member of the Poaceae family (Griffith et al. 1991). Lol p 1 differential expression was found in patients living in temperate and sub-tropic regions. Lol p 1 differential expression had occurred in allergic rhinitis participants in different biogeographical regions {temperate regions (latitude 35°S, 30.8 °C) and sub-tropic regions (latitudes 21 to 27°S, 27.6 °C)} (Kailaivasan et al. 2020). In our findings beta-expansin allergen expression in pollen reduces or diminishes with increase in temperature. Similarly, Fig. 4-II, Fig. 4-III and Fig. 5 denote Phl p 2/ Phl p 3 which are a 10.9 and 12 kDa pollen allergens (Dolecek et al. 1993; Petersen et al. 2006). These allergens contain a C-terminal expansin domain, and their sequence homology shows common origin of evolution with Phl p 1 (Petersen et al. 2006).

Phl p 2 and Phl p 3 expression differs in *P. pratense* pollen studied in the three regions. Phl p 2 and Phl p 3 expression is missing in R1 in western blot at 3 min exposure in UVP imager 8.5 while western blot 5 min exposure showed very slight expression of Phl p 2 in the R1 region. R1 has elevated temperature (23.06 °C) and decreased precipitation (42.31 mm) conditions as compared to R2 (22.07 °C, 102.445 mm) and R3 (21.39 °C, 110.34 mm) having low temperature and increased precipitation conditions. On molecular level, increasing temperature accumulates reactive oxygen species (ROS) and their role is important in pollen tube growth and causes pollen infertility (Pacini and Dolferus 2019; Santiago and Sharkey 2019). Proline amino acid levels decrease under heat stress conditions (Santiago and Sharkey 2019) which strengthens our findings where elevated temperature at R1 has reduced allergen expression. Similarly, the western blot 5 min exposure in UVP imager 8.5 showed very faint expression of Phl p 4 in R2 and R3

samples (see Fig. 5). Phl p 4 (55 kDa berberine bridge like-enzyme) occurs in exine, cytoplasm and amyloplast of pollen grain, and involved in the biosynthesis of sporopollenin (Fischer et al. 1996; Visez et al. 2021). It is a glycosylated protein and is a crossreactive carbohydrate determinant which produce low antibodies consequence to sensitization (Westman et al. 2020). Our results showed that increase in temperature has diminished or minimized the expression of Phl p 4. Similarly, Phl p 5 is a major and 32 kDa allergen. Till date, the allergen is not fully characterized and its role in pollen development and pollen release needs to be explored. Our results found that Phl p 5 expression was absent in R1 as compared to R2 and R3 (see Fig. 5). The study findings suggest that increase in air temperature and decrease in precipitation inhibit expression of the putative allergen in R1. Phl p 12 (14 kDa profilin) allergen is a ubiquitous pollen protein found in pollen grains of many monocot and dicot plants. The protein carries a high degree of sequence homology with other plant profilin (Valenta et al. 1994). Profilin regulates actin binding and cell signalling during cell division in pollen tube development (Jimenez-Lopez et al. 2012; Liu et al. 2021). We show that the allergenic profilin expression occurs in R2. Increase in temperature and decrease in precipitation inhibit the putative allergen expression in R1 region pollen. Differences in environmental conditions between the three regions affect the expression of Phl p 12.

Putative allergens Phl p 2 or Phl p 6 are expressed in R2 and R3 only (see Fig. 4-IV). Additionally, Phl p 12 and Phl p 5 showed expression in R3 only. Phl p 5 is an unknown allergen. Our findings suggest that Phl p 1 a cell wall protein (Hoffmann et al. 2022) is a major *P. pratense* pollen allergen (García-Mozo 2017) expression occurs in R2 and R3. Sensitization to *P. pratense* depends on the immunity level of a subject, allergen exposure time and allergen cross reactivity. Several allergens of grass pollen cause oral allergy syndrome (OAS) where allergy is induced by exposure of a subject to pollen allergens and food allergens (García-Mozo 2017). Such type of allergy due to allergen cross reactivity is caused by minor allergens like polcalcins, profilins and lipid transfer proteins (Radauer and Breiteneder 2006).

Conclusion

The study's findings elaborate how variations in climatic conditions influence *P. pratense*pollen size, biochemical composition and allergen expression. R1 region *P. pratense*-pollen were smaller in comparison to R2 and R3 region, and their respective FTIR spectral peaks marked differences in pollen biochemical composition. SDS PAGE showed loss of certain protein bands for the R1 sample in comparison to R2 and R3 samples. Similarly, western blot analysis showed that putative allergens-Phl p 1, Phl p 2 and Phl p 5 were absent in R1 pollen proteins, and were present in R2 and R3 pollen proteins. Additionally, we report for the first time *B. epidermidis and P. agglomerans* from *P. pratense* pollen through Biolog identification system that may have a role in *P. pratense*-pollen allergies which need to be further investigated. These findings suggest that soil pH conditions, elevation from sea level, increase in temperature up to 1 °C and decrease in precipitation may affect pollen biochemical composition and allergens expression during pollen development.

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Author contribution MH performed the experiment, compiled the data and wrote the manuscript. SN helped in project execution, data compilation and analysis and wrote the manuscript. RG helped with experimental design, co-supervised the work, reviewed the manuscript, and improved the English language of the manuscript. ZA conceptualized the idea, designed the experimental strategies, reviewed the manuscript and supervised the whole project.

Declarations

Data availability The data that support this study are available from the corresponding author upon appropriate request.

Conflict of interest The authors declare no conflict of interest.

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Broussonetia papyrifera Pollen Metabolome Insights, Allergenicity, and Dispersal in Response to Climate Change Variables

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Abstract: Background/Objectives: *Broussonetia papyrifera* is a tree-producing allergenic pollen that grows in varied climatic conditions worldwide and causes pollen allergies in susceptible humans. This study aimed to investigate *B. papyrifera* pollen morphology, pollen metabolome, pollen allergenicity, and climate change's impact on the plant habitat suitability in the future. **Methods**: Tree pollen was collected in spring from different regions of Pakistan. Pollen samples were subjected to morphological analysis, Fourier transform infrared spectroscopy (FTIR), liquid chromatography–mass spectrometry (LC-MS/MS), and immunoblotting. **Results**: MaxEnt modeling predicted the tree's future-growth invasion into new regions. Scanning electron microscopy (SEM) and FTIR displayed regional differences in pollen morphology and metabolome correlated to shifts in climatic variables. LC-MS/MS analysis detected four lipids that can potentially stimulate inflammatory responses. Pollen protein immunoblotting studies identified a putative 15 kDa novel allergen and verified previously known 40 kDa, 33 kDa, and 10 kDa allergens. *B. papyrifera* MaxEnt modeling through ACCESS1.0 and CCSM4 under 2-greenhouse gas emissions scenarios {representative concentration pathway (RCP) 4.5 and 8.5} projected the tree invasion by the years 2050 and 2070. **Conclusions**: The study findings demonstrate that differences in climatic variables affect *B. papyrifera*-pollen metabolome and predict the habitat suitability of the tree for invasion in the future. The study results provide a model system for studying other species' pollen morphology, metabolome, future habitat suitability for plant invasion, and associated allergies in response to climate change.

Keywords: *B. papyrifera* pollen; FTIR; LC-MS/MS; climate change; pollen allergy

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1. Introduction

Climate change is expected to have significant effects on plant growth and development and could impact pollen seasons [1]. The climate change conditions have affected plants [2]. Pollen, small in size, remains in the air for longer periods and is transported to longer distances [3,4]. Pollen has been widely studied in forensics, ecology, climate change, and aeroallergens investigations [5]. Aeroallergen research suggests an increase in the concentration of some allergenic pollen taxa and pollen allergies over the past 50 years [6]. This may be due to shifts in meteorological variables and an increase in air pollutants, which changed pollen quantity in the air in distinct bioclimatic regions [7]. An increase in temperature influences carbohydrates, proteins, and lipids levels in several pollen taxa [8].

Higher temperatures and elevated carbon and nitrogen concentrations affect pollen development [9]. Developing pollen requires a substantial amount of membrane lipids and fatty acids [10,11]. Temperature stress during pollen development affects pollen tube growth [12], pollen appearance, wall, and vesicular transport [13]. Increasing temperature causes an early bursting of the pollen tapetum layer, which may alter its allergenicity. The pollen is rich in mitochondria [14], and oxidative damage in the tapetum layer leads to increased production of reactive oxygen species (ROS), which causes programmed cell death [15]. ROS activation causes the regulation of pollen-associated lipid metabolites (PALMs) and phytoprostanes, which have been shown to evoke immune responses [16,17]. Oxylipins are involved in plants' growth and development and defend them from abiotic stresses [16]. Similarly, increasing ozone levels due to climate change have enhanced Birch pollen allergenic potential [18].

Broussonetia papyrifera is anemophilous (wind-pollinated), and its pollen is allergenic [19–21]. The pollen protein extracts analysis by skin prick test (SPT) and enzyme-linked immune sorbent assay (ELISA) identified 33-kDa and 40-kDa proteins as putative allergens that may elicit clinical responses in humans. The tree pollen is reported to cause urticaria and severe respiratory symptoms [22]. Sequence identities of the pollen IgE binding proteins have not been identified, and the tree produces large quantities of pollen of 12–17 µm size [23]. In Islamabad, Pakistan, *B. papyrifera*'s daily pollen count records show the highest concentration of 40,000 pollen grains/m^3 in the spring season [24]. According to the Global Diversity Information Facility (GBIF), *B. papyrifera* grows in Asia, in tropical and subtropical regions near the equator. This species was introduced in Pakistan in the early 1960s from East Asia for the beautification of Islamabad [20]. There is remarkable variation in climatic factors in different regions of Pakistan (average temperature and precipitation) [25,26].

Pollen exine morphology can be studied through microscopic techniques [27]. However, studying changes in pollen lipids, proteins, and carbohydrate constituents requires molecular approaches. Variations in the biochemical composition of pollen through Fourier transform infrared spectroscopy (FTIR) are informative for understanding the complex changes that can occur [28]. Kendel and Zimmermann have shown relative and absolute contents of lipids, carbohydrates, carotenoids, proteins, and sporopollenins. FTIR has been used as an easy and quick approach to study the biochemical composition of pollen where environmental stresses affect natural pollen samples [29,30]. Similarly, pollutants can affect pollen and intensify pollen allergies in humans [31]. Liquid chromatography–mass spectrometry (LC-MS/MS) can measure the presence of various chemical compounds in a sample by combining chromatography and mass spectrometry [32]. This study aims to find inter-regional variations in *B. papyrifera* pollen in varied climatic conditions, examine *B. papyrifera* pollen allergenicity, and predict the tree's future habitat suitability.

2. Materials and Methods

2.1. Study Area Characteristics

The geographical and climatic conditions of the study area are different, including district Peshawar Region 1 (R1), district Islamabad Region 2 (R2), and district Kotli Region 3 (R3). R1 is located at 331 m, R2 at 507 m, and R3 at 609 m elevation from sea level. The distance between R1 and R3 is 316 km; R1 and R2 is 187 km; and R2 and R3 is 132 km. The mean monthly temperature during spring for Region 1, Region 2, and Region 3 was 22.9 °C, 21.9 °C and 22.1 °C, respectively, and mean monthly precipitation during the period was 5.2 mm for R1, 59.5 mm for R2, and 21.1 mm for R3 [33]. *B. papyrifera* is allergenic and commonly grows in the three climatic regions. Regions R2 and R3 have lower average annual temperatures and increased mean annual precipitation than R1, which has increased

mean annual temperature and decreased mean annual precipitation. All study regions have variations in bioclimatic variables (Table S1, Figures S1 and S2).

2.2. *B. papyrifera* Growth Occurrence Record

The distribution of *B. papyrifera* in Asia was downloaded from the GBIF. Additionally, a field observation survey was carried out in *B. papyrifera* growing regions in Pakistan in 2020 and 2021 to record the plant occurrence. In the *B. papyrifera* growing regions survey, a minimum of 1 km of distant locations' longitudes and latitudes were recorded. However, the nearest accessible location was recorded for plants growing in the mountainous region, where recording the exact location was difficult. The presence of the tree was recorded through the Google Earth Pro version 7.3 mobile application. The field observation data combined with GBIF.org (accessed on 11 April 2022) data showed a record of 394 points in a dataset (See Table S2). The refined data were entered in a Microsoft Excel sheet for species with data (SWD) in CSV format to build the MaxEnt model (see Table S2).

2.3. Pollen Sampling

Mature pollen samples were collected from three *B. papyrifera* trees in each region growing at random spots in R1, R2, and R3 (Table S3, Figures S3 and S4) in the spring of 2020 and 2021. All sites were classified as green parks in urban habitats away from roads. Sterilized 50 mL tubes were used to store flowers to avoid pollen contamination from external sources. The samples were dried for 72 h in the shade at room temperature. Dried pollen grains were released from the samples by tipping the dry panicle of *B. papyrifera* on a 500 µm pore size mesh. The isolated pollen grains were then passed through a 100 µm pore size mesh and a 50 µm pore size mesh to collect pure pollen grains. The pure pollen samples were stored at room temperature (28 °C) in 1.5 mL sterile tubes.

2.4. Scanning Electron Microscopy of Pollen Samples

Scanning electron microscopy (SEM) was performed on dried *B. papyrifera* pollen samples on HITACHI SU 1500 equipment. Pollen was directly kept on SEM stub holders covered by a carbon patch. Pollen images were taken at 5 KV accelerating voltage in a high vacuum mode using SE detector at 50 µm resolution, and pollen diameters were calculated using the microscale tool of the SEM [34].

2.5. Fourier Transform Infrared Analysis of Pollen

Samples of 20 mg of dried pollen grains taken from each sample of *B. papyrifera* were placed on FTIR IRTracer-100 knob for analysis following the protocol as described by Kendel and Zimmerman [28]. Measurements were taken in triplicate with a Vertex 70 v FTIR Spectrometer using the attenuated total reflectance (ATR) method. Infrared radiations in the range of 500–4500 cm^{-1} and a resolution of 4 cm^{-1} were selected to draw peaks [35]. The peaks were normalized, and data files were imported in CSV format, which was processed to draw FTIR peaks through Origin Pro 8.5E-2018 software.

2.6. Liquid Chromatography–Mass Spectrometry Analysis of Pollen

Samples of dried pollen (100 mg) were placed in 15 mL tubes with 2 mL of methanol as an extraction fluid, and the tubes were kept in a shaker at room temperature for 1 h. The extracts were passed through a Solid Phase Extraction (SPE) assembly, and samples were collected in 2.0 mL tubes. Pollen extracts were subjected to LC-MS/MS analysis [36]. An Agilent triple quadrupole system was used for LC-MS/MS where compounds were separated (model: G1776A; dimensions: 160 mm × 435 mm × 436 mm; temperature: 40 °C). The injection volume of the sample was 5 µL. The mobile phase contained solvent A (100% acetonitrile) and solvent B (100% water). The mobile phase flow rate was 0.3 mL min^{-1}.

The HPLC gradient included 0–4 min. 2% A, 98% B; 4–7 min. 20% A, 80% B; 7–14 min. 90% A, 10% B; 15 min. 90% A, 10% B; 17 min. 2% A and 98% B. The ESI positive mode using nitrogen gas was used for mass spectrometry at 350 °C, with an 11 L min^{-1} gas flow, 4500 v capillary voltage, and r 50 psi nebulizer pressure to determine compounds. Identified compounds were compared with the National Institute of Standards and Technology (NIST)-17 database.

2.7. Pollen Protein Extracts Separated in SDS-PAGE Followed by Immunoblotting

Water-soluble pollen proteins were extracted in Bio-Rad™ 1X PBS buffer Cat # 1610763. Extracted proteins were quantified by Thermo Scientific NanoDrop 2000C and using a Cytiva™ 2D-Quant kit Cat # 80648354. Samples of 10 µg protein extract/well were separated in Novex 10–20% tris-glycine gels (Invitrogen™, Carlsbad, CA, USA, Cat # EC61355). A 3 µL sample of pre-stained Precision Plus molecular weight marker proteins was loaded (Bio-Rad, Hercules, CA, USA, Cat # 161-0374) on each gel to allow proteins apparent molecular weight estimation. Protein extract IgE binding performed using individual serum samples was confirmed by native immune-dot blot IgE binding activity of protein extracts with monoclonal mouse anti-human IgE conjugated with horseradish peroxidase (HRP) from Southern Biotech, Birmingham, AL (clone B3102E8 Cat # 9160-05). Human serum samples were purchased from Plasma Lab USA (Table S5). Allergens were confirmed through Western blotting and immune-dot blots with selected serum samples. Protein extracts were pre-heated at 25 °C, 70 °C, and 95 °C, cooled to room temperature, and then separated into premade gels (Novex). Proteins were transferred to 0.45 µm nitrocellulose membranes (Invitrogen™, Carlsbad, CA, USA, Cat # 645,239) and blocked with non-fat milk solution before applying diluted serum samples. Unbound IgE was removed by washing with 0.2% Tween 20 in PBS and incubated with Southern Biotech's monoclonal anti-IgE combined with horse radish peroxidase. The anti-IgE was visualized using Supersignal™ West Dura Extended Duration Chemiluminescent substrate (Pierce, Rockford, IL, USA, Cat # 34,076) in a UVP BioSpectrum 815 Imaging System for 1 min and 5 min as described by [37].

2.8. Temperature, Precipitation, and Spring Maximum Pollen Count Data Analysis

Mean monthly temperature and precipitation data from 2010 to 2020 for districts R1, R2, and R3 were retrieved from the Meteorological Department (PMD). Single day *B. papyrifera* maximum pollen concentrations per cubic meter during spring in R2 were retrieved from the PMD. GraphPad Prism 2.0 was used to apply two-way ANOVA on the data.

2.9. MaxEnt Modeling

We downloaded future climate data predictions of 19 bioclimatic variables (Table S6) from WorldClim 1.4 at 30-arc second resolution (0.93 × 0.93 km = 0.86 km^2 at the equator) [38] for ACCESS1.0 and CCSM4 models for the years 2050 and 2070 to representative concentration pathways (RCP), predicting urrence date in Asiad 4.5 and 8.5 climate change scenario [39]. The downloaded variable files were converted into ASC file format through QGIS 3.16.11. Bioclimatic variables were clipped onto the shapefile of Asia. These files were loaded on MaxEnt software version 3.4.4 and ran on the software [40]. Accuracy assessments for all models were randomly divided into 25% and 75% to conduct random tests and train the model. Background environmental data were applied by setting the background points maximum to 10,000 [41]. The model parameters were set to 5 replicates, and results were obtained from the average of all replicates of each model.

3. Results

SEM analysis showed tectate exines and horizontal apertures of pollen. Pollen diameter was 6–12 µm on 50 µm resolution. Pollen from different regions have variations in pollen diameter (Figure 1) and FTIR spectra (Figure 2). Pollen FTIR analysis showed variations in the protein (amide-II and amide-I) and lipid region. The environmental conditions of these regions varied in terms of mean annual temperature and mean annual precipitation (see Figures S1 and S2). The data analysis showed a gradual increase in mean annual temperature and a gradual decrease in mean annual precipitation across all regions. The LC-MS/MS analysis of pollen sampled in 2020 showed 33 different organic compounds. These compounds were from six organic groups. Alkaloids were abundant, terpenes, alkanes, and straight-chain fatty acids were the lowest in the pollen samples.

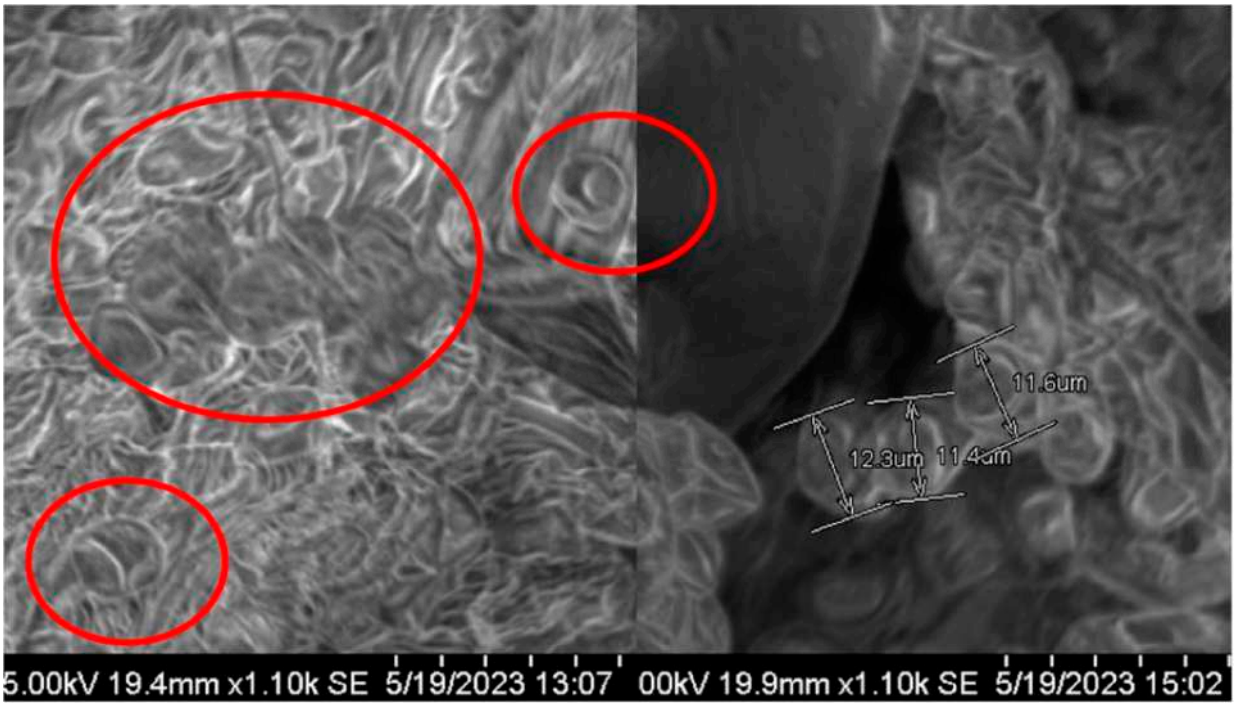

Figure 1. SEM images of *B. papyrifera* pollen captured at 50 µm resolution. Red circles indicate individual pollen, and double arrow lines show pollen diameter.

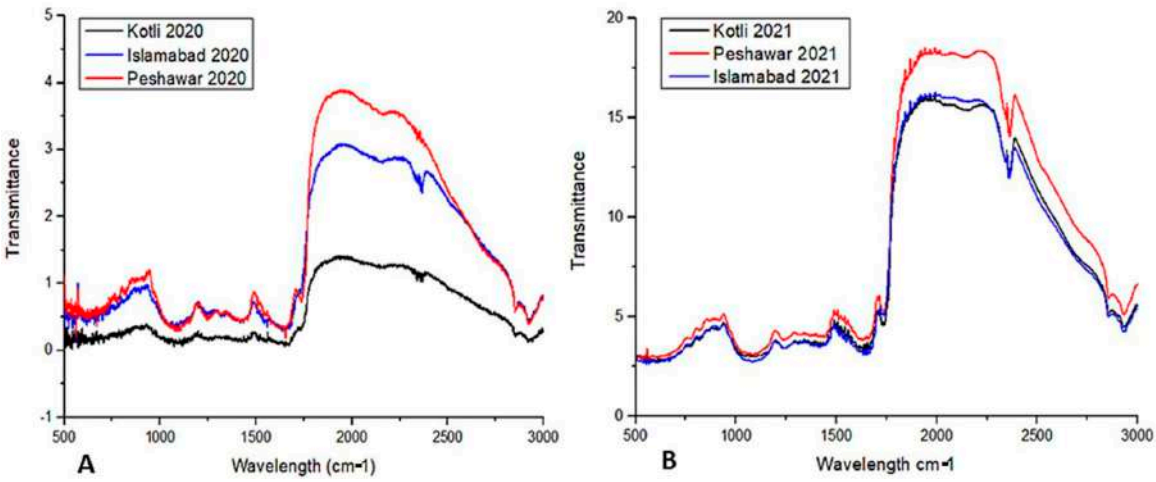

Figure 2. Inter-regional pollen FTIR analysis of *B. papyrifera*. Red, blue, and black lines denote R1, R2, and R3. (**A**) Pollen spectra of samples collected in the spring of 2020. (**B**) Pollen spectra of samples collected in the spring of 2021.

All pollen grains have tectate exine surfaces. R1, R2, and R3 pollen have diameters of 6–12 µm calculated through SEM imaging scale at 50 µm resolution. For each region, 50 pollen SEM images are averaged to show pollen grain diameter (Table 1).

Table 1. SEM data of pollen grain average diameter.

Image Name	Location and Time	Average Pollen Grain Diameter µm	Standard Deviation
i	Spring 2020 Peshawar (R1)	7.04	±0.61
ii	Spring 2020 Islamabad (R2)	8.86	±0.68
iii	Spring 2020 Kotli (R3)	10.02	±0.38
iv	Spring 2021 Peshawar (R1)	6.64	±0.36
v	Spring 2021 Islamabad (R2)	11.02	±0.62
vi	Spring 2021 Kotli (R3)	11.5	±0.26

3.1. Fourier Transform Infrared Spectroscopy (FTIR) of Pollen

FTIR analysis of pollen (Figure 2A) collected from R1, R2, and R3 in spring 2020 using a spectral wavelength of 500–3000 nm showed spectral regions of the protein (1700–1500 cm^{-1}) and lipid (2900–2700 cm^{-1}) were measured. The region between 1750 cm^{-1} and 2750 cm^{-1} exhibited greater differences than other spectral regions. Figure 2B shows FTIR spectra of pollen samples collected from R1, R2, and R3 in spring 2020 and 2021. The R1, R2, and R3 pollen spectra vary for percent transmittance values. Spring 2020 pollen spectra peak heights are different than spring 2021 pollen spectra peak heights. The spectral differences associated with chemical bonds for proteins and lipids are listed in Table 2 (see Table 2).

Table 2. Spectral zone, functional group frequencies, and associated chemical bonds.

Spectral Zone	Peak Frequency cm^{-1}	Chemical Bonds
Lipids	2930 2850	CH_3 stretching (lipids) CH_2 stretching (lipids)
Amide I 1700–1600 cm^{-1}	1620 β-sheet 1650 α-helix 1670 β-turn	NH_2 bending
Amide II 1600–1500 cm^{-1}	1550	N-O stretch

The table shows spectral zones of lipids and proteins along with peak frequencies and chemical bonds [31].

B. papyrifera pollen collected from natural environments was analyzed by FTIR, which showed differences in protein profiles for R1, R2, and R3 (Figure 3A,B). Spectral differences in peak heights occur in Amide-II and Amide-I functional groups (1500–1700 wavelength). Amide-II region spectral peaks differ for all sites pollen at 1550 cm^{-1} wavelength in transmittance for N-O stretch. Similarly, the spectral peaks differ at 1620 cm^{-1} β-sheet, 1650 α-helix, and 1670 β-turn of the NH_2 functional group in 2020 and 2021. R2 and R3 have sharp peaks N-O stretch at 1550 cm^{-1}, 1620 cm^{-1} β-sheet, 1650 α-helix, and 1670 β-turn (NH_2 functional group) in 2021 and 2020. The peaks for these wavelengths differ between spring 2021 and spring 2020 pollen for all sites.

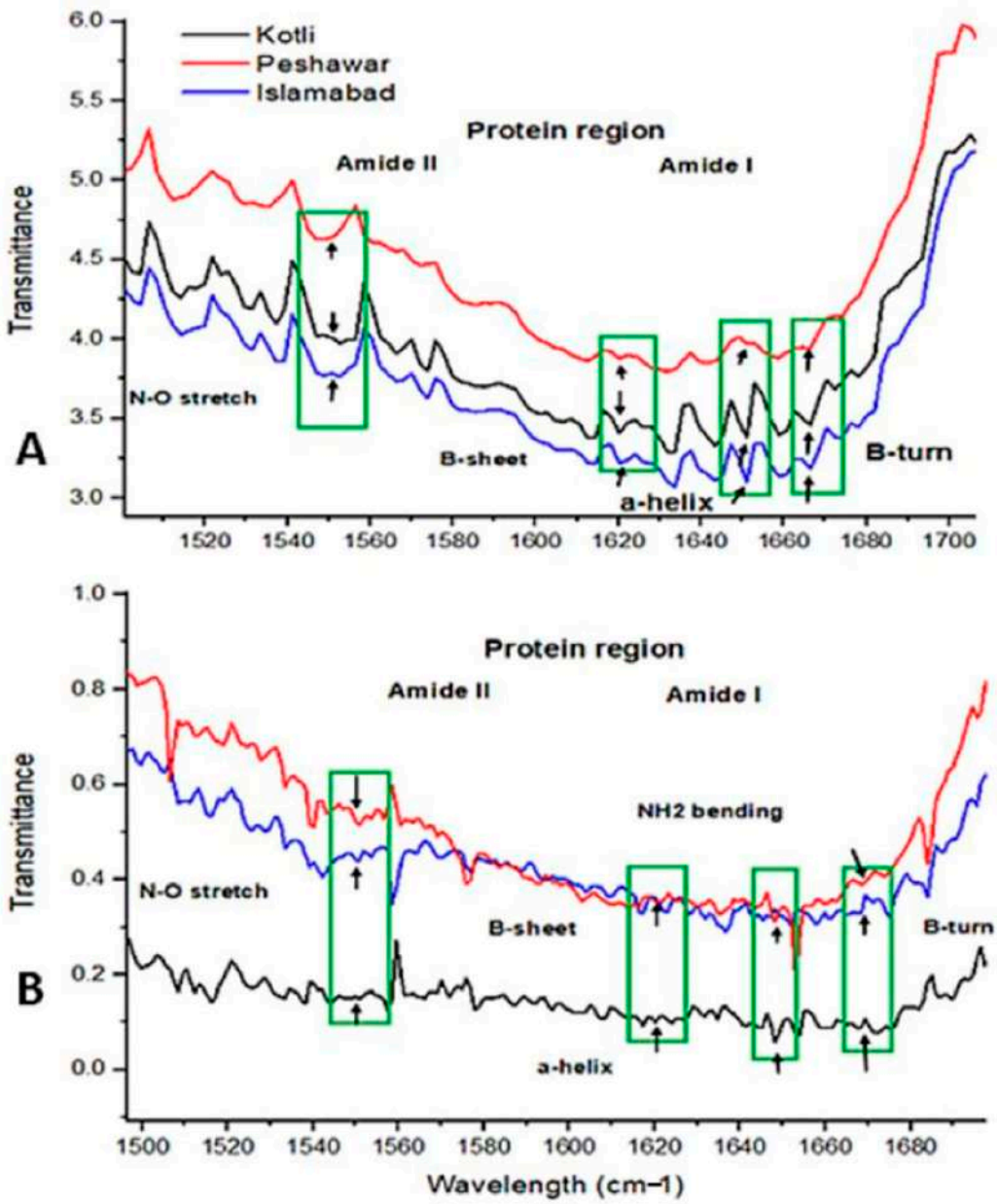

Figure 3. Location effect on FTIR spectra of pollen protein content collected in the spring 2020 and spring 2021: (**A**) Protein spectra of R1, R2, and R3 for pollen collected in 2020. (**B**) Protein spectra of R1, R2, and R3 pollen sampled in spring 2021. Green rectangles and black arrows show functional group comparisons.

The lipid region had variation in peak intensities in 2020 for CH_2 and CH_3 functional groups at 2850 cm^{-1} and 2930 cm^{-1} for R1, R2, and R3 (Figure 4A,B). Pollen in the same region lost peak intensities in 2021. These variations in spectral peaks reflect differences in climatic variables' impact on pollen structures.

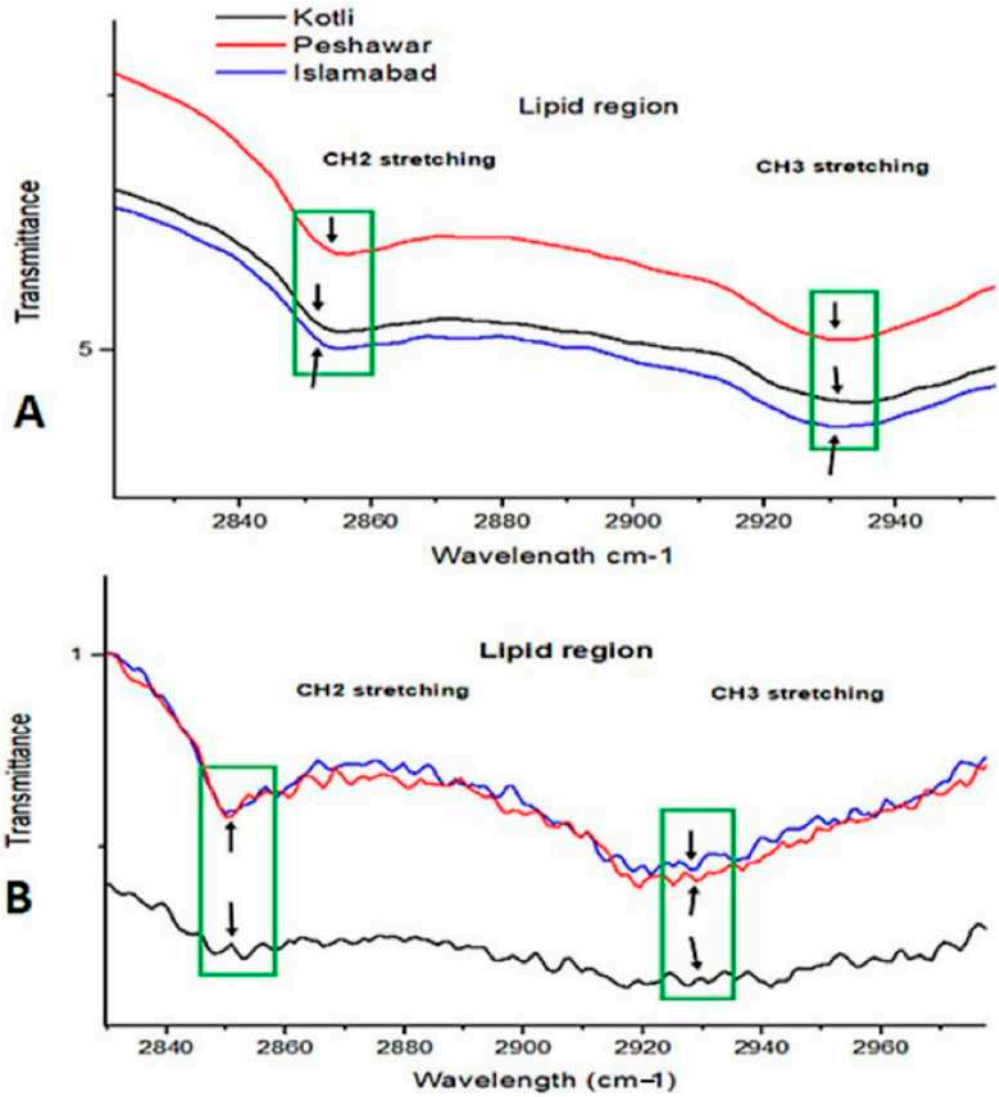

Figure 4. Seasonal and climatic effects on lipid region of pollen: (**A**) Spring 2020 pollen FTIR spectra for lipid region 2840 cm^{-1} to 2940 cm^{-1}. (**B**) Spring 2021 pollen FTIR spectra for lipid region 2840 cm^{-1} to 2940 cm^{-1}. Green rectangles and black arrows show functional group comparisons.

3.2. Light Chromatography–Mass Spectrometry (LC-MS/MS) Analysis

LC-MS/MS analysis of *B. papyrifera*-pollen run time was 17 min (Figure 5). LC-MS/MS analysis identified many compounds, as shown in Table S4. These compounds were categorized into seven groups, including alkaloids, carboxylic acids, alkanes, straight-chain fatty acids, terpenes, unsaturated fatty acids (USFA), and other lipids, as shown in Figure 6. Alkaloids occur in the highest numbers (12), followed by other lipids (9) and carboxylic acids (5). Alkanes, straight-chain fatty acids, and terpenes occur very little in quantity. There was only one straight-chain fatty acid ($C_{30}H_{52}O_2$), one alkene ($C_{21}H_{23}O_3P$), and one terpene ($C_{21}H_{26}N_2O_5$). Saturated fatty acids identified in this study include—Succinic acid, hexyl 2-phenylethyl ester ($C_{18}H_{26}O_4$), Adipic acid, di (2-phenylethyl) ester ($C_{22}H_{26}O_4$), Pimelic acid, di(phenethyl) ester ($C_{23}H_{28}O_4$), and Sebacic acid, 2-methylbenzyl undecyl ester ($C_{29}H_{48}O_4$) in the target pollen.

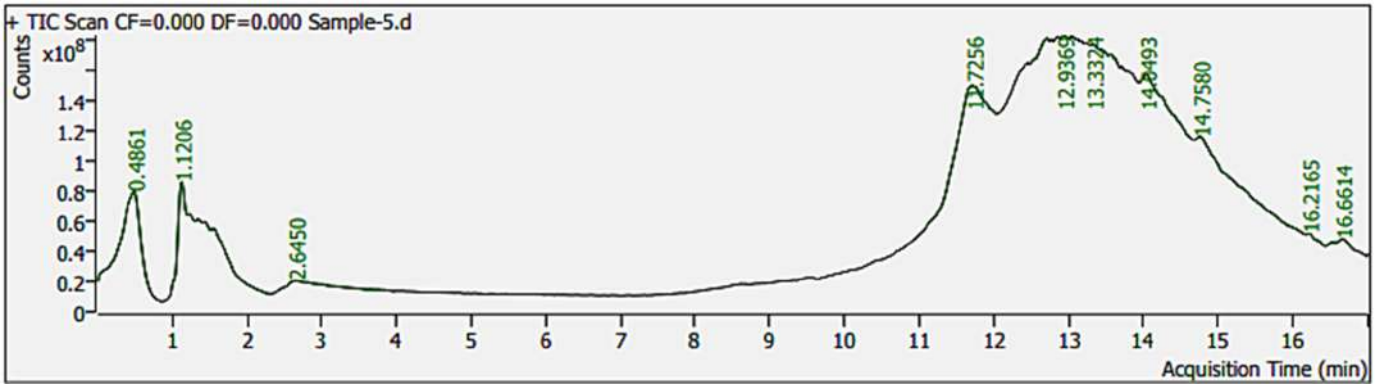

Figure 5. LC-MS/MS analysis of *B. papyrifera* pollen. The Y-axis shows charge to mass ratio for the ions of various compounds separated at different time intervals, while the X-axis denotes acquisition time in minutes. The sample run time is 17 min, with the highest peak at 13th minutes 12.9369. The analysis shows the variance in charge-to-mass ratios of ions in the pollen that make unique peaks. Peaks analysis reveals the presence of different compounds in the analyzed pollen (for details, see Table S4).

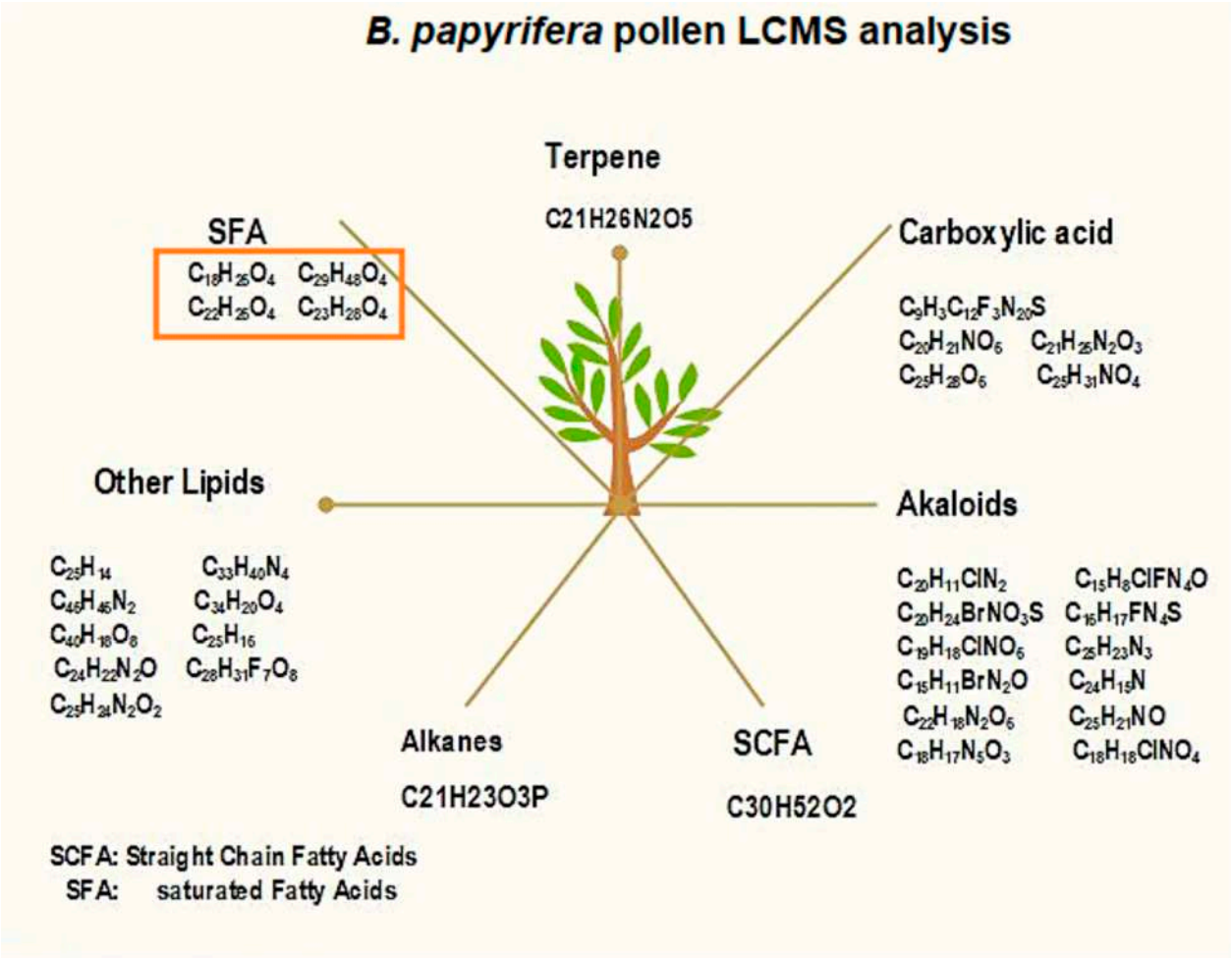

Figure 6. Schematic distribution of organic compounds in *B. papyrifera* pollen grains by LC-MS/MS analysis. Total compounds identified from the NIST database in the pollen samples were classified into 7 groups: alkanes, alkaloids, carboxylic acids, terpenes, other lipids, straight-chain fatty acids, and unsaturated fatty acids. Unsaturated fatty acids (marked in orange show the presence of allergenic compounds in the pollen grains of *B. papyrifera*.

3.3. In Vitro IgE Binding to *B. papyrifera* Pollen Proteins

Selected 13 serum 1:10 diluted samples in non-fat dry milk were cross-reacted with *B. papyrifera* pollen protein extracts in 1X PBS and 1:10 (1X PBS) diluted through dot blots. Three serum samples showed IgE binding with 1 µL pollen protein, and only one serum sample (32535-ET) showed IgE binding at 1:10 dilution. These three positive serum samples (Figure 7) were shortlisted for Western blotting to identify allergens in the pollen protein extracts.

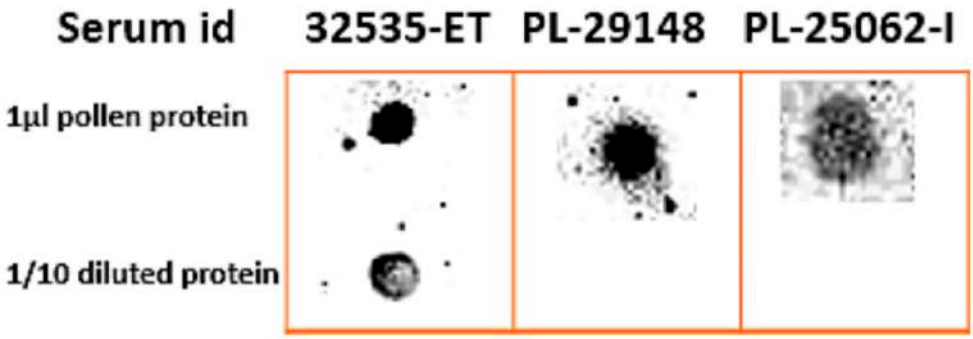

Figure 7. Dot blot of *B. papyrifera* pollen-protein extract. Samples of 1 µL of pure pollen-protein extracts and 1/10 diluted pollen-protein extracts in 1X PBS blotted on a nitrocellulose paper against 1:1000 diluted anti-IgE monoclonal HRP Southern Biotech antibodies. The dark spots show IgE binding with the respective serum ID given above the spot.

Allergenic proteins in the selected sera were determined through SDS gel electrophoresis and Western blotting. Western blotting identified four protein bands in lane 1 and lane 2 (see Figure 8). The band sizes 40 kDa, 33 kDa, 15 kDa, and 10 kDa were confirmed by comparison of the Western blot with the stained gel on the imager stage after Western blot visualization. Three protein bands of 40 kDa, 33 kDa, and 10 kDa were previously identified, while a 15 kDa protein band is not reported in any literature.

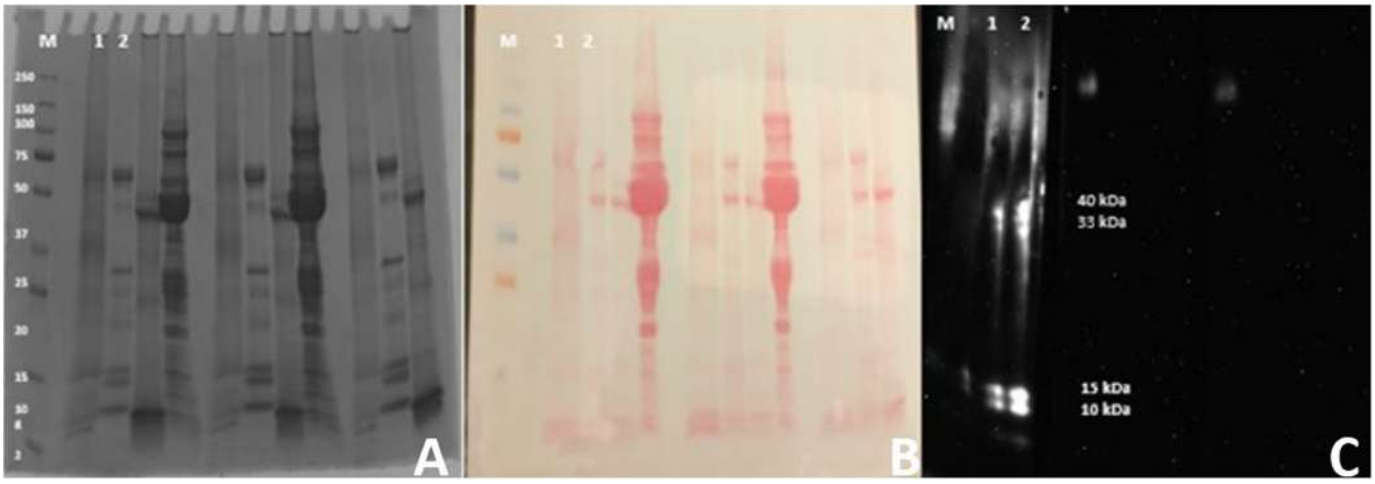

Figure 8. Western blot analysis of *B. papyrifera*-pollen protein: (**A**) *B. papyrifera*-pollen protein on a stained gel with 250 kDa pre-stained Precision Plus molecular weight marker proteins denoted as M, lane 1, and lane 2 shows *B. papyrifera*-pollen proteins. (**B**) Western blotting image of proteins on the nitrocellulose membrane stained with safranin red. (**C**) Anti-IgE binding with allergens on Western blot. The Western blot analysis depicts 4 allergen bands in lane 1 and lane 2, with primary antibody binding with 1:1000 diluted secondary antibodies.

3.4. Precipitation, Temperature and Pollen Count

The three decades' data analysis (1991–2020) for R1 and R2 and a single decade's data (2010–2020) available from PMD showed a decrease in mean annual precipitation and an increase in mean annual temperature during this period for the selected three regions (see Figures S1 and S2). The trend line for this period shows about 13 mm mean annual precipitation decrease for R1, while the trend line for R2 and R3 shows a few mm mean annual precipitation decrease. Similarly, the temperature change trend line for all three regions showed an increase in the temperature for the mentioned period. About 0.45 °C increase in mean annual temperature has occurred for R1. About 0.2 °C increase in mean annual temperature has occurred in R2 and R3. Analysis of spring maximum single-day pollen count in 1 cubic meter in R2 found pollen count production increase during 2003–2022 (see Figure S5). The maximum single-day pollen count was 38,946 pollen grains/m^3 in 2003, which reached the highest at 49,465 pollen grains/m^3 in 2022.

3.5. Model Performance of the MaxEnt Model

The area under the curve (AUC) is a built-in statistical package in MaxEnt that evaluates model performance based on the presence of records data only [42]. The AUC test exhibited 0.968, 0.99, 0.99, 0.938, 0.962, 0.987, 0.959, and 0.881 for training data of all models that showed excellent performance of the model in predicting the potential distribution of *B. papyrifera* in Asia in the years 2050 and 2070 under RCP 4.5 and RCP 8.5 (Figures 9 and 10).

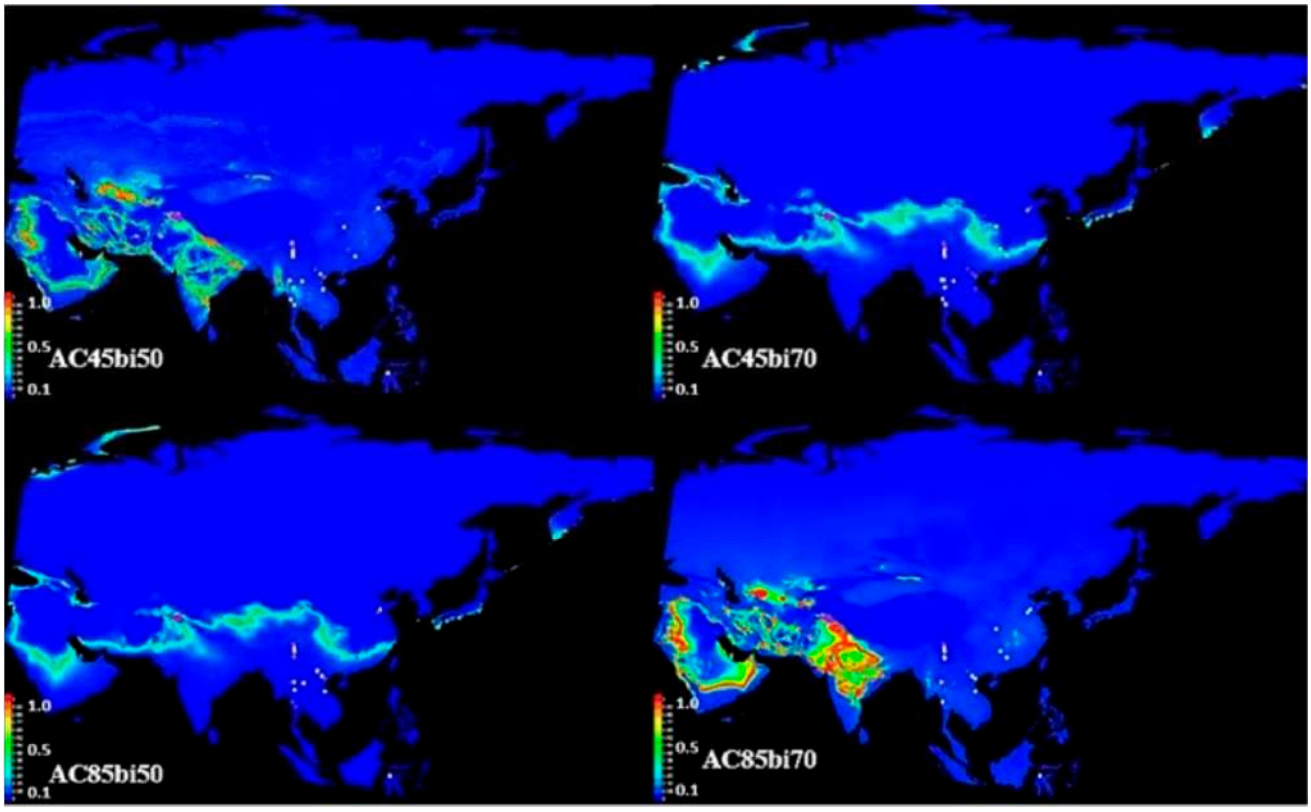

Figure 9. ACCESS1.0 based MaxEnt modeling of *B. papyrifera* growth occurrence in the years 2050 and 2070 in Asia. The figure shows habitat suitability for *B. papyrifera* in the years 2050 and 2070 through the ACCESS1.0 Coupled Model Intercomparison Project Phase 5 (CMIP5) dataset. AC represents ACCESS, 4.5 shows representative concentration pathway (RCP) 4.5, 8.5 shows representative concentration pathway (RCP) 8.5, and bi50 and bi70 denote bioclimatic variables in the year 2050 and 2070. The colored scale (0–1) on the right side of the map shows the suitability of locations for *B. papyrifera*-growth invasion in the future.

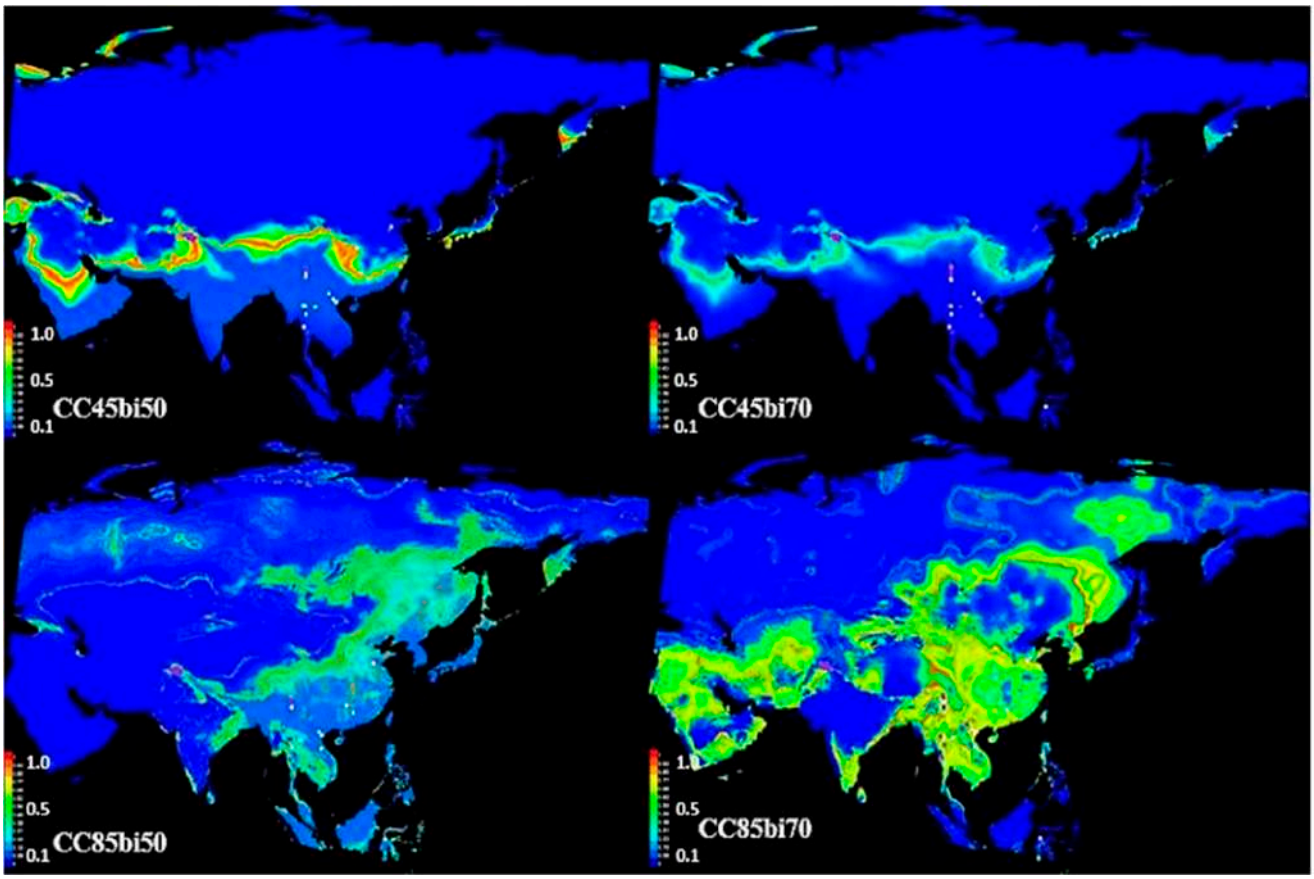

Figure 10. CCSM4-based MaxEnt modeling of *B. papyrifera* growth occurrence in the years 2050 and 2070 in Asia. The figure shows habitat suitability for *B. papyrifera* in the years 2050 and 2070 through the CCSM4 CMIP5 dataset. CC represents CCSM4, 4.5 shows representative concentration pathway (RCP) 4.5, 8.5 shows representative concentration pathway (RCP) 8.5, and bi50 and bi70 denote bioclimatic variables in the years 2050 and 2070. The colored scale (0–1) on the left side of the maps shows the suitability of locations for *B. papyrifera*-growth invasion in the future.

Out of 19 CMIP5 bioclimatic variables used in this study (Table S6), bioclimatic variables BIO701 (mean annual temperature), BIO7010 (mean temperature of warmest quarter), BIO7011 (mean temperature of coldest quarter), and BIO7012 (annual precipitation) have a significant contribution to the MaxEnt model analysis (Figure S6). The permutation importance of these four variables was the highest of all other variables that predict change in the values of these variables in the future will add to habitat suitability for plant growth. MaxEnt modeling through ACCESS1.0 for climate change scenario RCP 4.5 and RCP 8.5 shows a perspective increase in the occurrence of *B. papyrifera* in Asia. The increase occurs due to the mean annual temperature increase. AC4.5bi50 showed a greater increase in the possible habitat suitability of the tree in Saudi Arabia, Iraq, Jordan, Syria, Turkey, Iran, Afghanistan, Turkmenistan, Uzbekistan, Afghanistan, Pakistan, India, and Nepal, and a slight increase in China, Hong Kong, and Myanmar. The prospective invasion of *B. papyrifera* in model AC8.5bi50 is similar to AC4.5bi50, but the area covered in this model is greater, and the intensity of plant growth is also greater. Models AC4.5bi70 and AC8.5bi50, with almost similar temperature conditions, showed suitable growth conditions that can facilitate *B. papyrifera* invasion in regions of Saudi Arabia, Iran, Pakistan, India, Afghanistan, China, Nepal, Hong Kong, and Japan. Both RCP 4.5 and 8.5 scenarios in the years 2050 and 2070 show suitable habitats for *B. papyrifera* invasion into uninvaded regions of Asia.

Bioclimatic variables BIO701 (mean annual temperature), BIO7010 (mean temperature of warmest quarter), BIO7011 (mean temperature of coldest quarter), and BIO7012 (annual precipitation) contributed to the model's future assessment analysis. The model has assessed that a change in the values of these three variables in the future will improve the habitat for *B. papyrifera* growth, which will facilitate plant invasion and plant growth in new locations in Asia. MaxEnt modeling through CCSM4 for climate change scenarios RCP 4.5 and RCP 8.5 shows the prospective increase in the occurrence of the *B. papyrifera* tree. CC4.5bi50 shows habitat suitability where the plant can invade in the future. The CC4.5bi50 model assessment for the tree is similar to AC4.5bi50 and AC4.5bi70, with a slight difference in the invasion intensity. However, CC8.5bi50 showed a prospective habitat for the tree invasion in almost half of Asia in the year 2050, while the tree invasion will continue in Asia by 2070, according to CC8.5bi70. Both RCP 4.5 and 8.5 scenarios in the years 2050 and 2070 show the greater invasion of the tree into uninvaded regions in the years 2050 and 2070. Both models (Access 1.0 and CCSM4) predict habitat suitability for *B. papyrifera*, which can invade new locations in Asia in the years 2050 and 2070 climate change scenarios RCP 4.5 and RCP 8.5. However, the CCSM4 model predicts high habitat suitability for species invasion as compared to the ACCESS1.0 model in 2050 and 2070 under RCP 4.5 and RCP 8.5.

4. Discussion

The current research showed *B. papyrifera* pollen size/diameter (6–12 µm) through SEM analysis (Figure 1). Polysciences, a commercial pollen vendor, reported that the plant pollen size is 12–13 um, which is similar to our study's results [43]. SEM analysis did not find the varied temperature, precipitation, and elevation effects on *B. papyrifera* pollen's morphology. However, FTIR analysis showed differences in pollen biochemical composition through spectral peaks between 4000 cm^{-1} and 400 cm^{-1} (Figures 2–4). FTIR spectral peak differences can be due to changes in the functional groups of biomolecules making pollen. Pollen structure comprises lipids, carbohydrates, and proteins [44–48]. The *B. papyrifera* pollen FTIR analysis for protein and lipid functional groups showed differences in R1, R2, and R3 pollen grains. The spring 2020 pollen spectra varied in the protein region (1500 cm^{-1}–1700 cm^{-1}). Amide I region and Amide II region peaks differed at 1620 cm^{-1}, 1650 cm^{-1}, 1670 cm^{-1}, and 1550 cm^{-1} for β-sheet, α-helix, β-turn, and NO-stretch (Figure 3A,B). Peak areas in the Amide I region for β-sheet, α-Helix, and β-turn are similar for R2 and R3 and vary for R1. FTIR spectral peaks for CH_2 and CH_3 lipid functional groups vary for spring 2020 and 2021 pollens (Figure 4A,B). These FTIR results are in good agreement with the previous findings, which showed temperature and precipitation effects on pollen chemical composition and viability of *Syringa vulgaris* L. and *Phleum pratense* [28,33]. They found spectral differences in pollen of different species. Our findings show spectral differences in *B. papyrifera* pollen (protein and lipid region) collected from the three locations at varied temperatures and precipitation conditions. These interregional differences in the protein region of FTIR peaks can be correlated to variations in temperature and precipitation conditions of R1, R2, and R3 (Figures S1, S2 and 3).

The mean annual temperature of R1 is higher than R2 and R1, and the mean annual precipitation of R1 is lower than R2 and R3 (Figure S2). The spectra analyzed for the three regions showed elevated peaks and intense transmission for R1 as compared to R2 and R3. This enhancement in the spectra for R1 can be due to R1 bioclimatic variables and geographic location. Similar findings were made for hazel interregional pollen comparison through FTIR spectroscopy. Such variations in pollen biochemical structure are adaptive to bioclimatic variable differences [31]. High-temperature exposure reduces in vitro pollen germination percentage and pollen-tube length in *Pisum sativum* L. [12]. High-temperature

stress to pollen yields an increase in pollen proteins, while low-temperature stress causes enhancement of pollen lipids and carbohydrates [30,49]. Similarly, an increase in temperature and a decrease in precipitation affected pollen weight, protein content, and allergen contents [2]. High-temperature stress causes morpho-anatomical changes in pollen and its reproductive failure [10]. Comparably, FTIR spectroscopy identified adaptive variations in the chemical composition of proteins, lipids, and carbohydrates in pollen of *Poa alpina*, *Anthoxanthum odoratum*, and *Festuca ovina* grown in different geographic and climatic conditions [30]. Those investigations supported increases in temperature and elevation differences alter pollen chemical composition. Moreover, variations in environmental conditions have altered lipids, carbohydrates, and protein expression in 813 pollen specimens from 300 distinct plant species grown in five different pollen seasons [29].

Our investigations on *B. papyrifera* pollen' metabolome analysis through LC-MS/MS have found the presence of succinic acid, hexyl 2-phenylethyl ester ($C_{18}H_{26}O_4$); adipic acid, di (2-phenylethyl) ester ($C_{22}H_{26}O_4$); pimelic acid, di (phenethyl) ester ($C_{23}H_{28}O_4$); and sebacic acid, 2-methylbenzyl undecyl ester ($C_{29}H_{48}O_4$). Four lipid compounds (succinic acid, adipic acid, pimelic acid, and sebacic acid) have previously been reported to have a role in pollen allergies of some subjects [50,51]. These compounds' role was found in activating cytokine expression of DC/NKT cells and were marked as potential biomarkers [50]. These saturated fatty acids can induce T Helper 2 (TH2) cells associated with Interleukine-13 (IL-13) that produce stimulatory responses in humans. Exposure of epithelial layers of the nasal cavity to allergenic pollen (containing allergens and allergenic metabolites) elicits the human immune system.

This study reports four potential allergens in *B. papyrifera* pollen, causing pollen allergy in susceptible human populations. Previously, three allergens, 33 kDa, 40 kDa, and 10 kDa, were reported in *B. papyrifera* pollen through ELISA and blotting studies [20,52]. This study confirms the presence of 33 kDa, 40 kDa, and 10 kDa allergens in the *B. papyrifera* pollen and finds one novel allergen of about 15 kDa through Western blotting. Four *Moraceae* family allergens are recorded in Allergenonline.org (accessed on 7 January 2023) and Allergen.org (accessed on 7 January 2023) databases. Out of these four, Mor n 3 (non-specific lipid transfer protein) and two other 18 kDa unassigned are food allergens. Mor a 2 (cobalamin-independent methionine synthase) is the only registered airway allergen of 84 kDa in the allergen.org database. This study found a 15 kDa putative novel allergen and verified three previously identified airway allergens in *B. papyrifera* pollen. None of the four allergens identified in this study are registered in Allergenonline.org and Allergen.org. Historical pollen dataset analysis showed an increase in the spring single-day maximum pollen count trend (Figure S5). *B. papyrifera* pollen increase trend in the plant growing regions makes the susceptible population more vulnerable to pollen allergy. It is expected that *B. papyrifera* pollen allergy will increase in the future due to high habitat suitability for plant growth and distribution.

RCP 4.5 estimates a moderate increase in global temperature and carbon availability in the atmosphere [53]. MaxEnt modeling, according to RCP 4.5, estimates greater habitat suitability for *B. papyrifera* growth in new regions of Asia (Figure 9). Similarly, an extreme increase in global temperature and carbon emissions, according to RCP 8.5 [54], assessed habitat suitability through MaxEnt modeling, which suggests *B. papyrifera* invasion into uninvaded regions of Asia in the future (Figure 10). ACCESS1.0 and CCSM4 models through Maxent estimated habitat suitability for the invasion of the allergenic species into new locations. Further, analysis of bioclimatic variables [mean annual temperature (BIO701), mean temperature of warmest quarter (BIO7010), mean temperature of coldest quarter (BIO7011), and BIO7012 (annual precipitation)] percent contribution and permutation to the two models assessed their role in *B. papyrifera* invasion. An increase in BIO1, BIO10,

and BIO11 variable values in the future due to climate change contributes to the model output. The model output correlated the tree growth habitat suitability with mean annual precipitation and temperature. Similarly, changes in the pollen metabolome correlated with changes in mean annual temperature and precipitation. An increase in these variables values in the future occurs due to increase in carbon emissions [55]. The expected increase in the habitat suitability for *B. papyrifera* growth can facilitate its invasion into new regions. *B. papyrifera* invasion will expose the susceptible human population to the allergenic pollen that may invoke immune reactions. Similarly, ragweed pollen allergy 2041–2060 assessment in Europe found that the plant invasion will cause pollen sensitization in 33 to 77 million people [56].

5. Conclusions

Our state-of-the-art investigations have found a putative novel airway allergen (about 15 kDa) in *B. papyrifera* pollen through Western blotting and verified the presence of three other allergens reported earlier. So far, none of these four airway allergens are registered in Allergenonline.org and Allergen.org. The study's findings demonstrate that factors like temperature, precipitation, and differential elevation may have affected *B. papyrifera* pollen biochemical composition in the R1, R2, and R3 regions. Among 33 organic compounds identified by LC-MS/MS analysis, four saturated fatty acid compounds have a possible role in eliciting cytokine expression of DC/NKT cells in the human immune system. *B. papyrifera* MaxEnt modeling estimated the habitat suitability for the plant growth in Asia in the years 2050 and 2070 can enhance the plant pollen allergy. The study findings provide a baseline model system for studying other allergenic plant species' pollen metabolome, predicting their growth invasion, and associated allergies in response to climate shift.

Supplementary Materials: The following supporting information can be downloaded at: https://www.mdpi.com/article/10.3390/metabo15020137/s1, Figure S1. Mean annual precipitation of R1, R2, and R3 from 2010 to 2020 and their two-way ANOVA. Black, red, and green colors represent R1, R2, and R3. The line graph shows the mean annual precipitation, and the bar graph shows their two-way ANOVA. Figure S2. Mean annual temperature of R1, R2, and R3 from 2010 to 2020 and their two-way ANOVA. Black, red, and green colors represent R1, R2, and R3. The line graph shows the mean annual temperature, and the bar graph shows their two-way ANOVA. Figure S3. Map of Pakistan showing *B. papyrifera* pollen sampling regions. The map shows regions R1, R2, and R3, where *B. papyrifera* pollen was sampled in spring 2020 and spring 2021. Pollen was collected from three nearby trees from each region. Figure S4. *B. papyrifera* pollen collection in the study regions. Figure S5. *B. papyrifera* pollen count from 2002 to 2022. Figure S6. Individual bioclimatic variables contribution to *B. papyrifera* MaxEnt modeling. Table S1. Mean annual precipitation and mean annual temperature data were obtained from PMD through email requests. Table S2. *B. papyrifera* occurrence in Asia sheet in csv file format. The Excel sheet contains *B. papyrifera* occurrence in Asia obtained from GBIF.org and field observation in Pakistan. Table S3. *B. papyrifera* sampling sheet. The sheet contains pollen sampling information from R1, R2, and R3 in spring 2020 and spring 2021. Table S4. *B. papyrifera* pollen LC-MS/MS analysis results. Table S5. Sera purchased from PLASMA Lab USA in August–November 2022 allergic to *B. papyrifera* pollen. Table S6. Bioclimatic variables downloaded from Worldclim.org at 30 s resolutions.

Author Contributions: M.H. performed the experiment, compiled the data, and wrote the manuscript. S.N. helped in project execution, data compilation, and analysis and wrote the manuscript. Z.A. conceptualized the idea, designed the experimental strategies, reviewed the manuscript, and supervised the whole project. R.E.G. helped with experimental design, co-supervised the work, reviewed the manuscript, and improved the English language of the manuscript. All authors have read and agreed to the published version of the manuscript.

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ORIGINAL PAPER

Correlation of high pollen concentrations with air pollutants and meteorological parameters: "the case of Islamabad, Pakistan"

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Abstract

Air pollution influences climate change and contributes to respiratory allergies, yet interactions between these factors are less understood. The escalation of air pollutants and temperature increases can influence pollen concentrations, shifts in pollen seasons, and an increase in seasonal allergies. This study investigates trends of *Broussonetia papyrifera*, *Pinus*, *Cannabis sativa*, and grasses (Poaceae) pollen in Pakistan over the past 16 years, as well as their correlations with air pollutants and meteorological parameters. Minute-based data of 10 air pollutants were recorded using a HAZ-6000 scanner in Islamabad during 2022-23. Datasets on meteorological parameters (such as precipitation, temperature, wind speed, and relative humidity) and pollen concentrations (2008–2023) were retrieved from the Pakistan Meteorological Department (PMD). Both datasets were normalized, and monthly averages were calculated. Spearman correlation of pollen concentrations with air pollutants and meteorological factors was performed. Mean monthly pollen concentrations, meteorological parameters, and air pollutants were plotted utilizing the NCAR Command Language (NCL). Interestingly, O_3, CO_2, and NO_2 showed a positive correlation, whereas SO_2, NO, and CO showed a negative correlation with daily pollen concentrations/m^3. Pollen season trends comparative analysis indicated a shift in seasonal pollen integral (SPIn), start date, and duration. This is the first study from Pakistan to suggest that increasing temperature and air pollutants correlate with higher pollen concentrations and extended pollen season. This research provides a baseline system for examining allergenic pollen season shifts over time, indicating an increase in pollen allergy abundance with climate change in South Asia.

Keywords Pollen season shift · Climate change · Air pollution · Respiratory allergies · South asia

Introduction

Allergenic pollen exposure causes allergies and asthma in susceptible individuals (Taketomi et al. 2006; Lam et al. 2024). Pollen allergies are rising with climate change (Frenguelli 2014; Ziska 2021). Growing atmospheric pollution is correlated to increasing pollen allergy burden, temperature increase, and climate change (Ziska 2020; Manisalidis et al. 2020; Almetwally et al. 2020; Eguiluz-Gracia et al. 2020; Alzate-Marin et al. 2021; Rentschler and Leonova 2023; Verscheure et al. 2023). Atmospheric pollution levels have increased (Rentschler and Leonova 2023; Li et al. 2023) with the global population increase and expansion of the city scale (Lu et al. 2021). Atmospheric pollutants differ from region to region, depending on the population and development (Kuerban et al. 2020). Non-biological agents, nitrogen oxides (NO_x), carbon monoxide (CO), carbon dioxide (CO_2), sulfur dioxide (SO_2), particulate matter $PM_{2.5}$, PM_{10}, and volatile organic compounds (VOCs) concentrations in the air are rising (Albertine et al. 2014; Ziska 2020; Almetwally et al. 2020; Alzate-Marin et al. 2021). Biological components in the air, such as pollen, spores, bacteria, and viruses, can be affected by air pollutants (Hoebeke et al. 2018; Lo et al. 2019; Gehrig and Clot 2021; Damialis et al. 2021; Plaza et al. 2022; Markey et al. 2022; Esterhuizen

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et al. 2023; Ortega-Rosas et al. 2023). Air pollutants and allergenic pollen in the air affect human health. Anthropogenic air pollution causes up to 9 million deaths annually (Manisalidis et al. 2020). Air quality monitoring and regulation, as outlined by the World Health Organization (WHO) and the Environmental Protection Agency (EPA), are critical for maintaining a healthy life. Increasing population and decreasing land cover in urban areas (Kamran et al. 2023) pollute the air quality.

Pollen and pollutants are monitored globally (Oteros et al. 2019; Ziska 2020; Almetwally et al. 2020; De Weger et al. 2021; Bilal et al. 2021; Plaza et al. 2022). Air quality data from the monitoring stations show that Asia's air quality is poorer in comparison to Europe and America (Kuerban et al. 2020). South Asia and China constitute densely populated and polluted parts of the world (Rentschler and Leonova 2023), and bear an unhealthy atmosphere (Bilal et al. 2021; Sharma and Mauzerall 2022; Taghizadeh et al. 2023; Torabi et al. 2023; Yousaf et al. 2024). Rigorous air quality monitoring of the region is obligatory to assess how regional air quality affects the global allergy burden. In Pakistan, air pollutants in Karachi and Lahore exceed recommended healthy levels (Bilal et al. 2021; Nawaz 2022; Khan et al. 2023b; Yousaf et al. 2024). However, the effects of mounting air pollutants on pollen allergies are less studied. This research investigates air pollutants, shifts in pollen seasons, and pollen correlations with air pollutants and meteorological parameters in Islamabad.

Materials and methods

Study area

The study was conducted in Islamabad, an urban area and the capital city of Pakistan (Fig. 1). The city lies on the border with another densely populated city, Rawalpindi. The total area of Islamabad is 906.84 km^2, and Rawalpindi is 259 $km^{2,}$ while the total population of Islamabad is 1.4 million and Rawalpindi is 2.4 million (Mannan et al. 2021). The urban area lies 507 m above sea level at the slope of the Margalla hills at 33.49 N and 72.24 E latitude and longitude. The southern and western regions of the study area have lower elevations, while the northern and eastern regions have higher elevations. Construction on bare land in the study area has increased from 5.7% (52 km^2) to 25.7% (233 km^2) and 3.7% (60 km^2) to 14.1% (228 km^2) from 1990 to 2021 in Islamabad and Rawalpindi, respectively (Kamran et al. 2023). Automobiles, construction, and other small industries release air pollutants in the study region. Allergenic plant species like *Broussonetia papyrifera*, *Phleum pratense*, *Pinus* and *Cannabis sativa* densely grow in the study area. Wind blows mostly southwest, which disperses pollen in the air, and frequent rainfall in the target region naturally cleans the atmosphere. The topography of the study area forms an air basin and retains pollen and air pollutants at lower elevations due to slow wind speed. Pollen and pollutant dispersion are faster at higher elevations due to high wind speed. Furthermore, the temperature inversion in the air basin traps pollen and air pollutants at lower elevations and hinders their vertical dispersion.

Data collection

HAZ-SCANNER™ model HIM-6000 was installed ~ 5 m above ground level and 100 m away from road at COMSATS University Islamabad, Pakistan, at 33.6518° N and 73.1566° E (Fig. 1). The scanner was programmed to sample and log every minute data for ozone (O_3), nitrogen oxides (NO, NO_2), sulfur dioxide (SO_2), carbon dioxide (CO_2), carbon monoxide (CO), VOCs, methane (CH_4), and polymethyl acrylate (PMA) since January 2nd 2022 to November 1 st 2023. Daily average values were calculated from the data (See S2 table). Additionally, temperature, precipitation, relative humidity, windspeed, and pollen datasets of *B. papyrifera*, *Pinus*, *C. sativa*, and Poaceae were taken from the Pakistan Meteorological Department (PMD) for 2008–2023 (See S1 table). PMD has a Burkard Hirst-type pollen monitoring system installed on a rooftop ~ 10 m above ground level in H-8/2 Islamabad at 33.6836°N and 73.0639°E, which manually monitors daily pollen concentration/m^3 in the air (Fig. 1). In addition, PMD recorded observational data for weather parameters, including temperature, relative humidity, and precipitation. Airborne pollen concentrations/m^3 data were extracted from the spring and summer pollen season dataset. According to the Koppen climate classification, spring and summer seasons occur from March to April and May to August, respectively (Kottek et al. 2006). Air pollutants data from 2022 to 2023 were correlated with airborne pollen season concentration data for the respective period, see sheet S2 in the Supplementary file.

Data standardization

Minute-based data was obtained for 10 different pollutants {CO2, CO, NO_2, NO, CH_4, O_3, PMA, SO_2 and VOCs} from the HAZ-6000 scanner. The data obtained for CO_2, CO, and CH_4 were in parts per million (PPM) and were converted into mg/m^3 by using Formula (1) NO_2, NO, O_3, and SO_2 data were in parts per billion (PPB) and were converted into µg/m^3 by using Formula (2) The data for PMA was in µg/m^3 standard form, and the data for VOCs was used in PPB standard form. For pollen correlation analysis and season shift, the pollutants' minute-based dataset was converted into

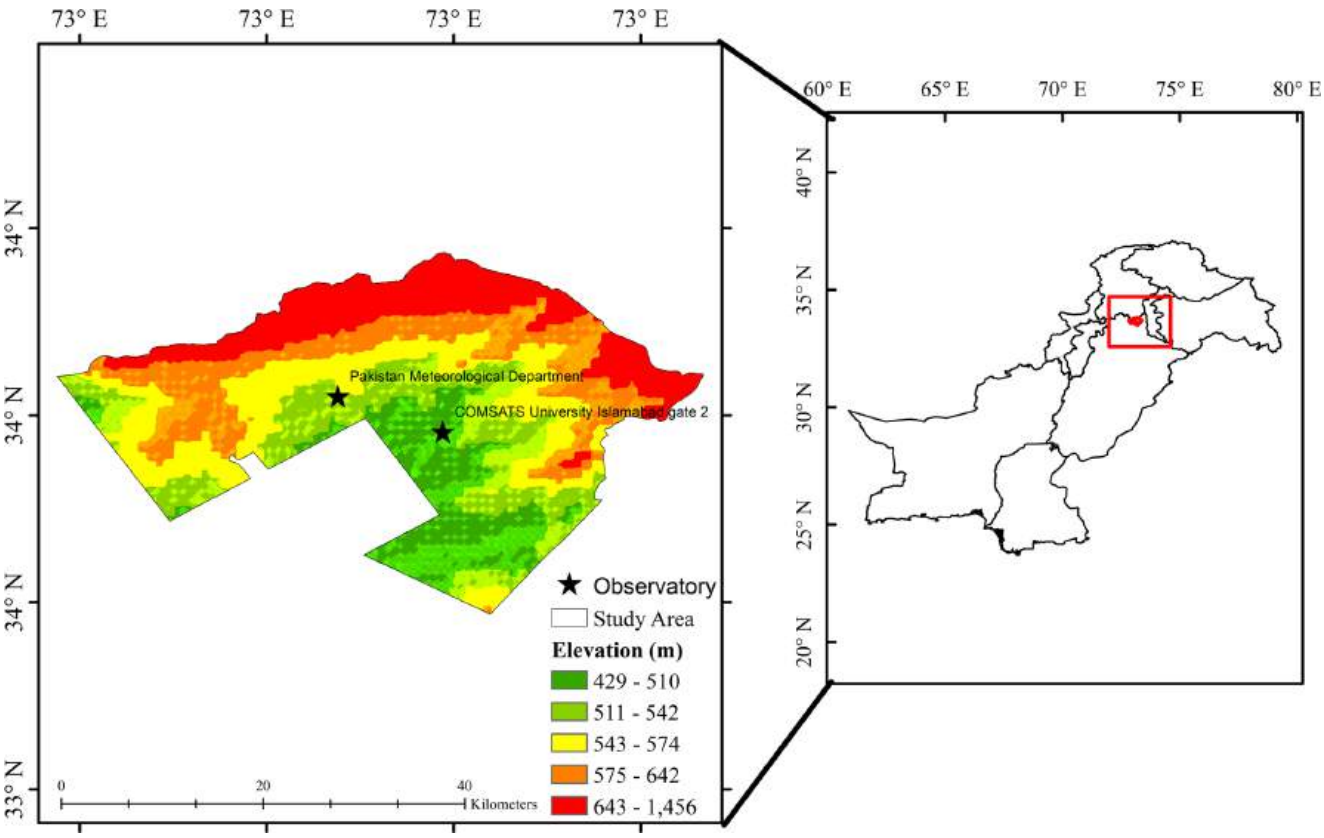

Fig. 1 Geographical map of the study region. Figure 1 shows a map of Pakistan, and the insert map shows the study region. The compass rose depicts directions, the Y-axis shows latitude, the X-axis shows longitude, and the scale bar represents visual distances on the map. Stars on the map denote data observatories. Different colors mark elevations of the study area. The figure was plotted using ArcMap Version 10.8 https://desktop.arcgis.com/en/arcmap/latest/get-started/setup/arcgis-desktop-system-requirements.htm

daily averages by using Microsoft Excel 2016 functions. Similarly, the pollen concentration, precipitation, temperature, and relative humidity dataset was converted into daily averages by using Microsoft Excel 2016 functions.

$$\frac{mg}{m3} = PPM \times \ molecular\ weight \div 24.45 \times density\ of\ air \div molar\ volume \quad (1)$$

$$\frac{mg}{m3} = PPM \times molecular\ weight \div 24.45 \times density\ of\ air \div molar\ volume \times 1000 \quad (2)$$

Meteorological data and pollen season determination

Grass and birch pollen studies indicated that the warmer season was related to the pollen season onset (Adams-Groom et al. 2022). Pollen season was determined by using the definition "Start at the first day of a sequence of 3 days with at least 5 pollen/m^3 and end at the last day of such a sequence" (Bastl et al. 2015). It is observed that the *B. papyrifera* pollen season starts with concentrations higher than 1000/m^3. At the end of the pollen season, a high number of pollen grains are released from the detached catkins for several days after flowering has stopped. Therefore, *B. papyrifera* airborne pollen concentrations higher than 1000 pollen/m^3 for 3 consecutive days were considered as the pollen season start. Airborne pollen concentrations less than 1000 pollen/m^3 for 3 consecutive days were considered as the end of the season. Each year, tails data was added up to the end of the pollen season and outlier data was excluded from the season comparative analysis. Daily Tmeans within each period were averaged to calculate mean seasonal temperatures. Similarly, SPIn was calculated according to the Bastl et al. (definition 1 in the manuscript).

Pollen trend and spearman correlation analysis

Five pollen-trend variables for all pollen were investigated: season onset, SPIn, pollen high number days (HN), duration of the season, and the first high day (FH) for all pollen for 16 years accordingly (Adams-Groom et al. 2022). A high day is classed as a daily average per cubic meter of air of 50+for Poaceae (Koivikko et al. 1986), 5+, 5+and 1000+for *C.*

sativa, *Pinus* and *B. papyrifera*, respectively. Spearman correlation analysis was used to investigate the daily average pollen season concentration correlation with meteorological parameters and air pollutants since January 2nd, 2022, to November 1 st, 2023 (See S2 Table). Graphs were made for means using Excel functions and were compared with the WHO and the National Environmental Quality (NEQ) Standards of Pakistan (Khwaja et al. 2020).

Results

Monthly pollen concentrations, temperature, precipitation, relative humidity, and windspeed in spring/summer in 2022 and 2023

The results from Burkhard Hirst's type pollen monitoring system showed that pollen concentrations of *B. papyrifera*, *Pinus*, *C. sativa*, and Poaceae were high in the air during the pollination period. *B. papyrifera* pollen concentration was highest, 49,434 pollen/m^3 in spring, in comparison to summer, 3800 pollen/m^3. *Pinus* and Poaceae pollen concentrations were highest in spring, were 175 and 65 pollen/m^3, while *C. sativa* pollen concentration was highest in summer, 100 pollen/m^3 (Fig. 2). The daily temperature, rainfall, and relative humidity from March to August in 2022 and 2023 varied (Fig. 2). The wind speed in Islamabad ranged between 0 and 10 m and blew mostly in a south-west direction (Fig. 3).

Air pollutant concentrations

Daily average analysis of the dataset showed high levels of all pollutants in the air during pollen season in 2022-23. Daily average air concentrations of O_3, NO, PMA, CO_2, and CO (Fig. 4 - b, c, f, g, and i) were higher in 2022, while CH_4, SO_2, VOCs, and NO_2 (Fig. 4 - a, d, e, and h) were higher in 2023. These air pollutants' minimum and maximum concentrations during pollen season are listed in Table 1.

Statistical correlation between pollen concentration and meteorological parameters/air pollutants

Spearman correlation coefficient at alpha 0.05 shows that *B. papyrifera* pollen concentrations were significantly and positively correlated with CO_2, CO, O_3 and NO_2, and negatively correlated with monthly mean rainfall, average temperature increase, and average NO and SO2 increase. *Pinus* pollen/m^3 at alpha 0.05 was significantly and positively correlated with increasing NO_2 and O_3 concentrations and negatively correlated with an increase in monthly total rainfall, monthly mean humidity, temperature, CO_2, CO, NO, and SO_2. *C. sativa* pollen/m^3 have a p-value of 0.02, and were positively correlated with relative humidity, temperature, PMA, and SO_2 increase, and negatively correlated with increasing monthly total rainfall, CO_2, CO, NO_2, O_3, and PMA. Grass pollen/m^3 at alpha 0.05 were significantly and positively correlated with NO_2 and O_3, and negatively correlated with increasing CO, NO_2, NO, SO_2, VOCS, and PMA concentrations (Fig. 5, S2 Table).

Pollen season trend analysis

Pollen season trend analysis illustrates a shift in pollen season from 2008 to 2023. SPIn of *B. papyrifera* pollen has increased, from 516,798 to 641,097 pollen/m^3. *Pinus*, Poaceae, and *C. sativa* SPIn have decreased from 1178, 2573, and 3901 to 553, 765, and 1919 pollen/m^3, respectively (Fig. 6). *B. papyrifera*, *Pinus*, Poaceae and *C. sativa* pollen season start dates in 2008 have shifted from 8th March, 6th March, 10th February, and 26th March to 3rd March, 27th February, 13 February, and 4th March, respectively, in 2023 (S1 table). *B. papyrifera* and *C. sativa* and *Pinus* pollen seasons have extended, and the Poaceae pollen season is unchanged.

B. papyrifera average daily pollen concentrations comparative analysis showed that the pollen season started on 8th March in 2008 started earlier on 26th February in 2021. Additionally, 45,000 pollen/m^3 were recorded in 2008, which increased to about 49,000 pollen/m^3. Pollen season and pollen/m^3 have extended (Fig. 7).

Pinus daily average pollen comparative analysis from 2008 to 2023 showed that a slight shift of the early pollen season has occurred, while the season end is almost similar. Similarly, the maximum pollen/m^3 recorded almost remained analogous except for 2019, where about 240/m^3 were recorded (Fig. 8).

C. sativa daily average pollen/m^3 comparison showed that *C. sativa* pollen season started around 27th March in 2008 and ended around 30th October, while in 2023 the pollen season onset was around 4th March and ended around 30th October. The maximum average pollen/m^3 recorded in a single day has decreased from 120 pollen/m^3 in 2008 to 80 pollen/m^3 (Fig. 9).

Grass pollen average daily pollen/m^3 analysis showed that the pollen season start and end dates remained somewhat similar. However, a decrease in single-day maximum pollen concentrations from 140 to 90 pollen/m^3 has occurred, except for 2020, where 260 pollen/m^3 were recorded (Fig. 10).

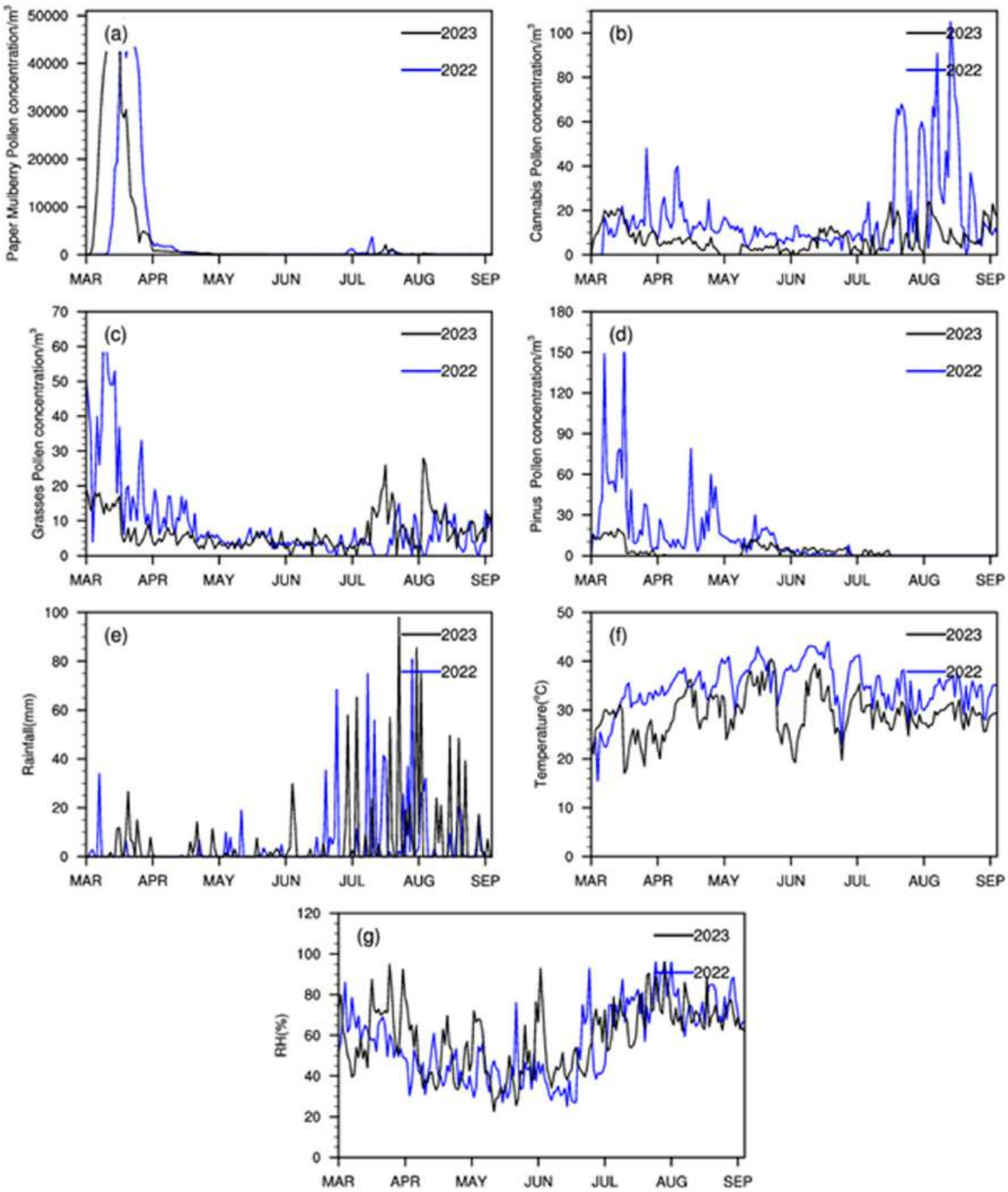

Fig. 2 Pollen concentration, monthly temperature, relative humidity (%), and precipitation in Islamabad, Pakistan, from March to August 2022 and 2023. The Y-axis shows pollen/m^3 and the X-axis shows the time scale (**a**) *B. papyrifera* pollen (**b**) *C. sativa* pollen (**c**) Poaceae pollen (**d**) *Pinus* pollen concentration (**e**) mean monthly rainfall, (**f**) temperature, and (**g**) relative humidity

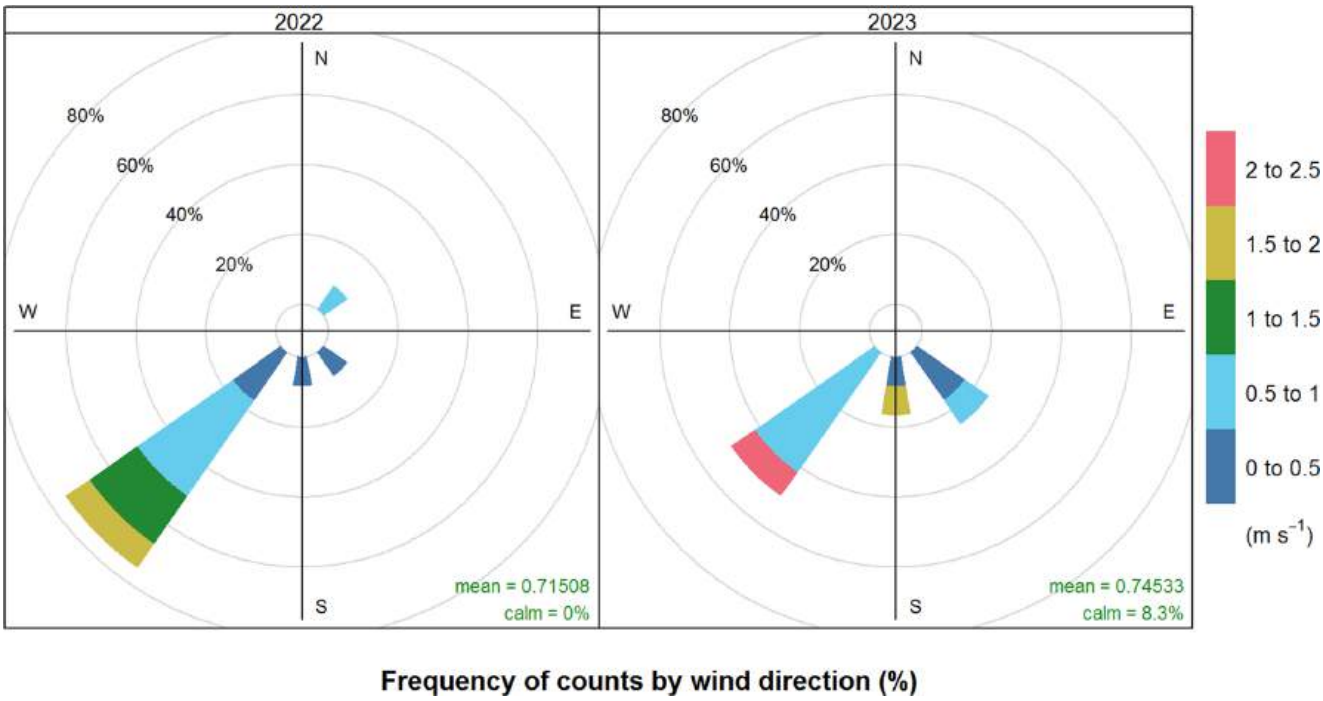

Fig. 3 Average monthly wind speed in Islamabad during March-August in 2022 and 2023. The color marker represents wind speed in m/s^{-1}, and the petals show wind direction. The percentage indicates the frequency of wind direction

Discussion

Air pollutants act as environmental stressors, significantly influencing plant biomass and biochemical processes. For example, elevated levels of carbon dioxide (CO_2) and nitrogen dioxide (NO_2) have been associated with increased pollen production and heightened allergenicity, respectively (Verscheure et al. 2023; Ziska 2020). Additionally, higher concentrations of ozone (O_3) and nitrogen oxides in the atmosphere can accelerate pollen release, potentially altering the duration of the pollen season. Our study found that NO_2 concentrations peaked during the winter and spring seasons of 2022 and 2023, exceeding 30 µg/m³, while the lowest levels were observed in summer, remaining above 10 µg/m³ (see Fig. 4-i). Similarly, O_3 and CO_2 levels were also highest during winter and spring. CO_2 reached its maximum in December 2022, with values just below 850 mg/m³ (see Fig. 4-h). Carbon monoxide (CO) concentrations were consistently above 0.6 mg/m³ during the study period, further contributing to air quality degradation (see Fig. 4-j).

Spearman correlation analysis between air pollutants and total pollen concentration revealed significant positive correlations for CO_2, CO, NO_2, and O_3 (see S2 Table). Elevated CO_2 levels, which support photosynthesis, contribute to increased plant biomass and, consequently, higher pollen production. Although methane (CH_4) concentrations peaked at 850 µg/m³ (see Fig. 4-a), no significant correlation was observed between CH_4 and total pollen concentration. In contrast, increases in atmospheric concentrations of sulfur dioxide (SO_2), nitric oxide (NO), and VOCs were negatively correlated with total pollen levels. NO concentrations were highest in August and July - reaching 80 µg/m³ in 2022 and 44.31 µg/m³ in 2023 - and lowest in February and April, with values of 2.69 µg/m³ (2022) and 4.38 µg/m³ (2023) (see Fig. 4-d). All observed pollutant concentrations in Islamabad exceed the WHO 2021 Air Quality Guidelines (AQGs). These findings are consistent with pollutant trends observed in Lahore, where post-COVID-19 levels of PM_{10}, SO_2, NO_2, NO, and O_3 increased by 270%, 100%, 115%, 300%, and 70%, respectively. The rise in pollution is likely attributable to increased vehicular traffic and industrial emissions. Similar trends were reported by Nawaz et al. in a comparative study of air quality in the megacities of Lahore and Karachi, which also concluded that air pollution levels in both cities exceed WHO-recommended standards (Nawaz 2022). The WHO recommends that ambient concentrations of key air pollutants remain below the following thresholds: PM_{10} – 15 µg/m³, NO_2 – 10 µg/m³, SO_2 – 15 µg/m³, NO – 15 µg/m³, O_3 – 15 µg/m³, and CO – 4 µg/m³ (Ji et al. 2022). In this study, the highest pollen concentrations were recorded during two distinct periods: February to April and August to September. These seasonal peaks are consistent with previous findings from Rawalpindi, a neighboring city of Islamabad (Naseer et al. 2024).

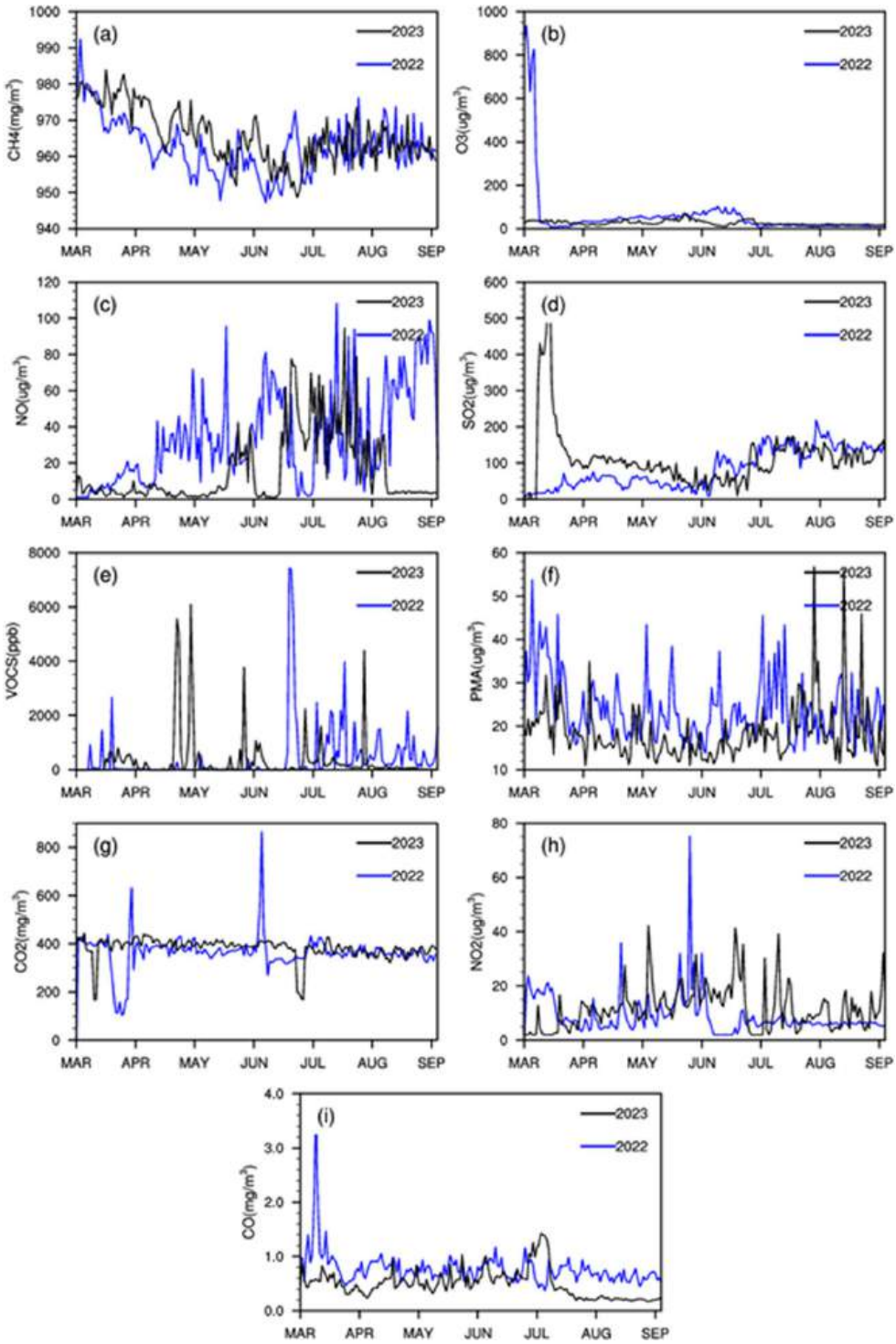

Fig. 4 Mean monthly air pollutant values of Islamabad in 2022-23. The Y-axis shows the concentration of the respective pollutants, and the X-axis shows the time scale. **a**, **b**, **c**, **d**, **e**, **f**, **g**, **h** and **i** show CH_4, O_3, NO, SO_2, VOCS, PMA, CO_2, NO_2, and CO concentrations in air/m^3

O_3 and NO_2 concentrations correlations with pollen

Our study found that average ozone (O_3) concentrations were significantly and positively correlated with the airborne pollen concentrations of *B. papyrifera*, *Pinus* spp., and members of the *Poaceae* family (see Fig. 5). During the pollination seasons of these species, O_3 levels ranged between 30 and 80 µg/m³ (see Fig. 4-C). Similarly, nitrogen dioxide (NO_2) concentrations, which ranged from 8 to 40 µg/m³ (see Fig. 4-i), were positively associated with pollen concentrations of *B. papyrifera*, *Pinus*, and *Poaceae*, but negatively correlated with *C. sativa* pollen levels (see Fig. 5). These findings are supported by recent research from Daelemans, who reported that nitrogen

Table 1 Minimum and maximum single-day air pollutant concentrations in 2022-23

Pollutant	Minimum	Year/month	Maximum	Year/month
CH_4	950.16 mg/m^3	June 2022	992.13 mg/m^3	March 2022
O_3	9.72 µg/m^3	July 2023	935.48 µg/m^3	March 2022
NO	1.00 µg/m^3	July 2022	81 µg/m^3	May 2023
SO_2	0.12 µg/m^3	June 2022	573.54 µg/m^3	March 2023
VOCs	0.85 PBB	April 2023	7424 PPB	July 2022
PMA	0.77 µg/m^3	April 2023	53.73 µg/m^3	August 2023
CO_2	0.67 mg/m^3	March 2023	1542.15 mg/m^3	June 2022
NO_2	1.19 µg/m^3	March 2023	42.58 µg/m^3	January 2022
CO	0.32 mg/m^3	August 2023	3.83 mg/m^3 and	March 2022

enrichment can increase grass pollen abundance and allergenicity (Daelemans 2025).

High relative humidity levels above 60% have been shown to facilitate the uptake of atmospheric pollutants by pollens (Galveias 2021). High concentrations of air pollutants can affect pollen development, alter morphology, and increase allergenic potential. For example, *Phleum pratense* L. pollen exposed to nitrogen dioxide (NO_2) in a reactor acidified the pollen surface which was directly correlated with increased allergenicity in humans (Chassard et al. 2015). The finding highlights the impact of pollutant uptake on pollen physiology, biochemistry and allergenicity. In our study, increasing concentrations of nitric oxide (NO) (see Fig. 4-d) were negatively correlated with airborne pollen levels of all studied species (see Fig. 5). This is consistent with previous findings showing that exposure to nitrogen oxides adversely affects pollen viability in several species, including *Betula pendula*, *Carpinus betulus*, *Ostrya carpinifolia*, *Cercis siliquastrum*, *Quercus mongolica*, and *Pinus nigra* (Verscheure et al. 2023). Interestingly, *Lolium perenne* pollen collected in Madrid, Spain, under elevated concentrations of NO_2 and SO_2, exhibited overexpression of glycoside hydrolase family 17, proteins with EF-hand motifs, pectinesterases, LTPs, and thaumatin (Lucas et al. 2020), indicating a stress response at the molecular level. Similarly, Sčevková et al. reported that exposure to atmospheric NO_2 can increase the concentration of the major allergen Phlp5 per pollen grain, further highlighting the role of air pollutants in enhancing pollen allergenicity (Ščevková et al. 2020).

Pinus radiata pollen collected from a polluted environment showed up-regulation of three stress-related genes (García-Gallardo et al. 2012). In addition to changes in protein expression, the adsorption of nitric acid (HNO_3) and nitrous acid (HNO_2) on the pollen surface leads to

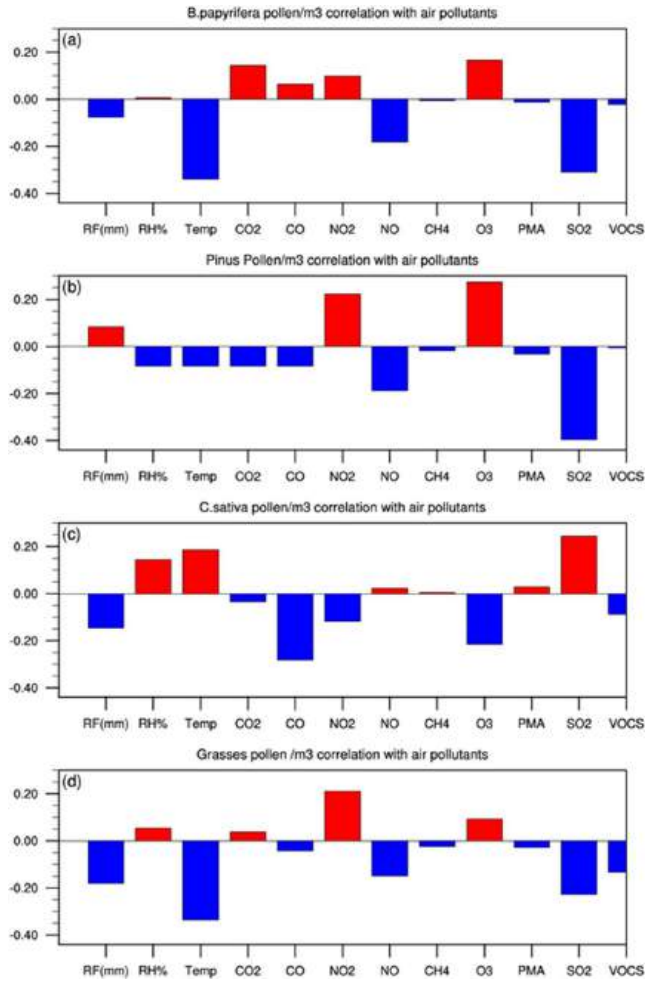

Fig. 5 Pollen concentration Spearman correlations with meteorological factors and air pollutants. Alphabets on the left side of each correlation matrix show individual pollen species correlations. The Y-axis indicates the correlation value and the X-axis indicates individual factors. Bars on the X-axis show pollen correlation with air pollutants. **a***B. papyrifera* pollen concentration correlation with air pollutants. **b***Pinus* pollen concentration correlation with air pollutants. **c***C. sativa* pollen concentration correlation with air pollutants. **d** Grass pollen concentration correlation with air pollutants

acidification, which can irritate the nasal mucosa and contribute to pollinosis (Verscheure et al. 2023). Elevated nitrogen pollutant levels have also been associated with increased expression of allergenic proteins in pollen. Our findings indicate that mean monthly NO_2 concentrations exceeded 10 µg/m³ in March 2022 and 20 µg/m³ in March 2023 (see Fig. 4-i). Based on these observations, we suggest that rising NO_2 levels—and possibly other atmospheric pollutants may play a synergistic role in exacerbating springtime pollen allergies in the Islamabad region. Supporting this, Ravindra et al. reported a significant positive correlation between NO_2 and both annual and seasonal pollen concentrations of *Celtis occidentalis*, while *Morus alba* and *Eucalyptus* species showed no significant correlation

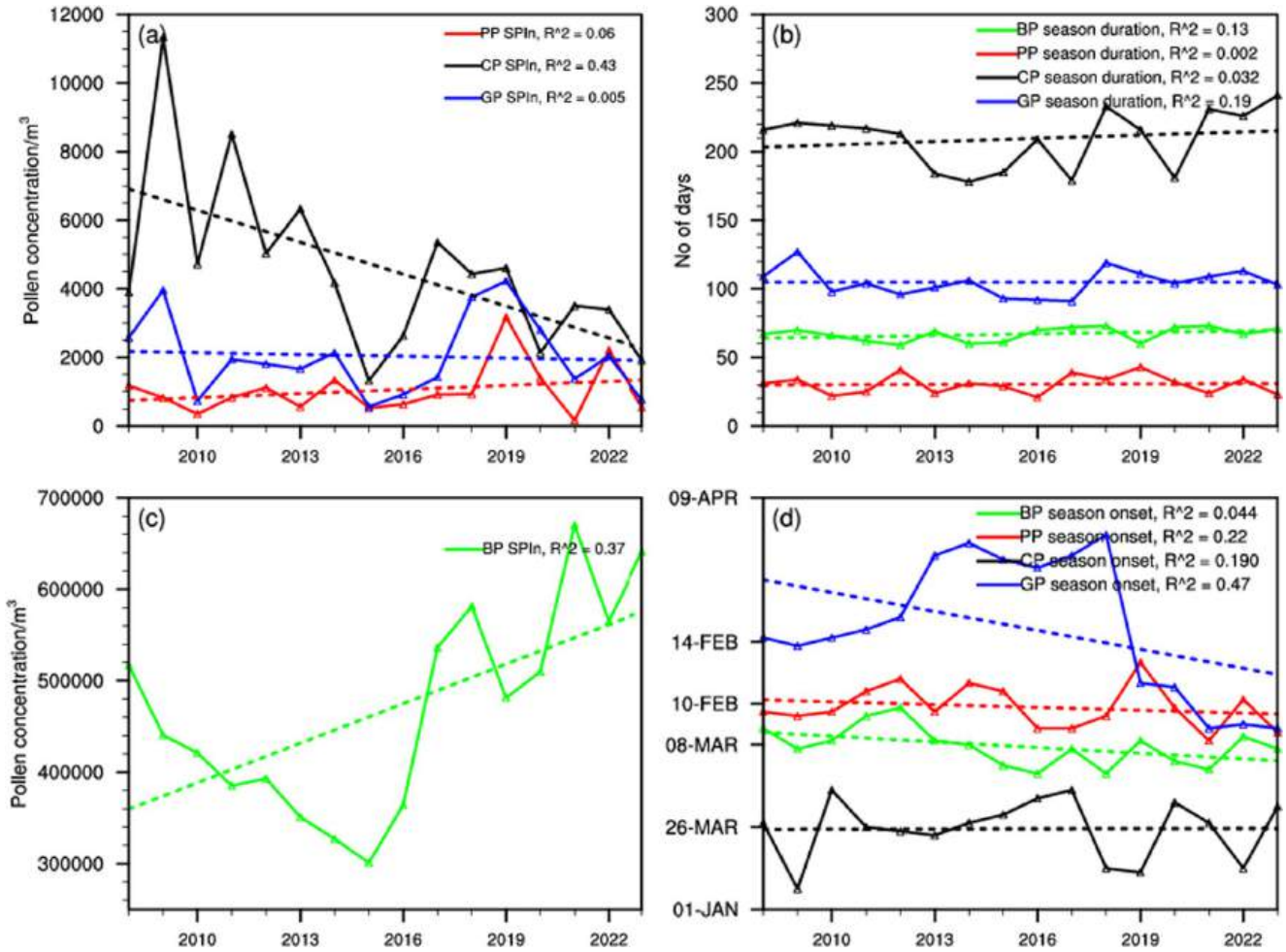

Fig. 6 Pollen season trend of ***B. papyrifera*** (BP), ***C. sativa*** (CP), ***Pinus*** (PP), and Poaceae (GP). Different colors mark pollen species types. Colorful dotted lines indicate trends. Sub-figures **a** and **c** show the seasonal pollen integrals. Sub-figure **b** indicates pollen season duration. Sub-figure **d** indicates pollen season start dates and pollen concentration trend. The X-axis shows the yearly time scale in all subfigures. The Y-axis shows pollen concentration in sub-figures **a** and **c**, and the number of days in sub-figure **b**. The Y-axis shows the pollen season start date in sub-figure **d**

(Ravindra et al. 2021). Additionally, NO_2 has been shown to increase pollen allergenicity and fragility (Verscheure et al. 2023). Long-term data also support the impact of air pollution on pollen production. An analysis of an 18-42 year dataset from 17 pollen monitoring stations demonstrated a significant increase in pollen production at 14 locations, attributed to rising CO_2 concentrations (Ziska 2020).

Carbon emissions' impact on pollen

CO_2 is a critical component of photosynthesis and plays a significant role in plant growth and pollen production. The current study found that increased concentrations of CO_2 and carbon monoxide (CO) were positively correlated with higher *B. papyrifera* pollen levels, while negatively correlated with *Pinus*, *C. sativa*, and Poaceae pollen concentrations (Fig. 5). During 2022 - 2023, ambient CO_2 levels generally ranged between 400 and 425 mg/m³, with notable peaks observed in March and June 2022 (Fig. 4h). Notably, *C. sativa* typically pollinates in late spring through summer, coinciding with periods of lower CO_2 concentrations. These findings align with previous research. For instance, Damialis et al. reported a doubling of atmospheric pollen concentrations per decade across 12 out of 16 monitored plant species in Greece (Damialis et al. 2007). Furthermore, the Intergovernmental Panel on Climate Change (IPCC) Fifth Assessment Report (Working Group II) highlighted that climate change may enhance the production, release, and dispersion of airborne pollen and fungal spores, contributing to increased incidence and severity of allergic diseases such as asthma, allergic rhinitis, conjunctivitis, and dermatitis (Damialis et al. 2021). *Betula papyrifera* at elevated CO_2 concentrations showed an increase in pollen production (Darbah et al. 2008). Extended warmer seasons,

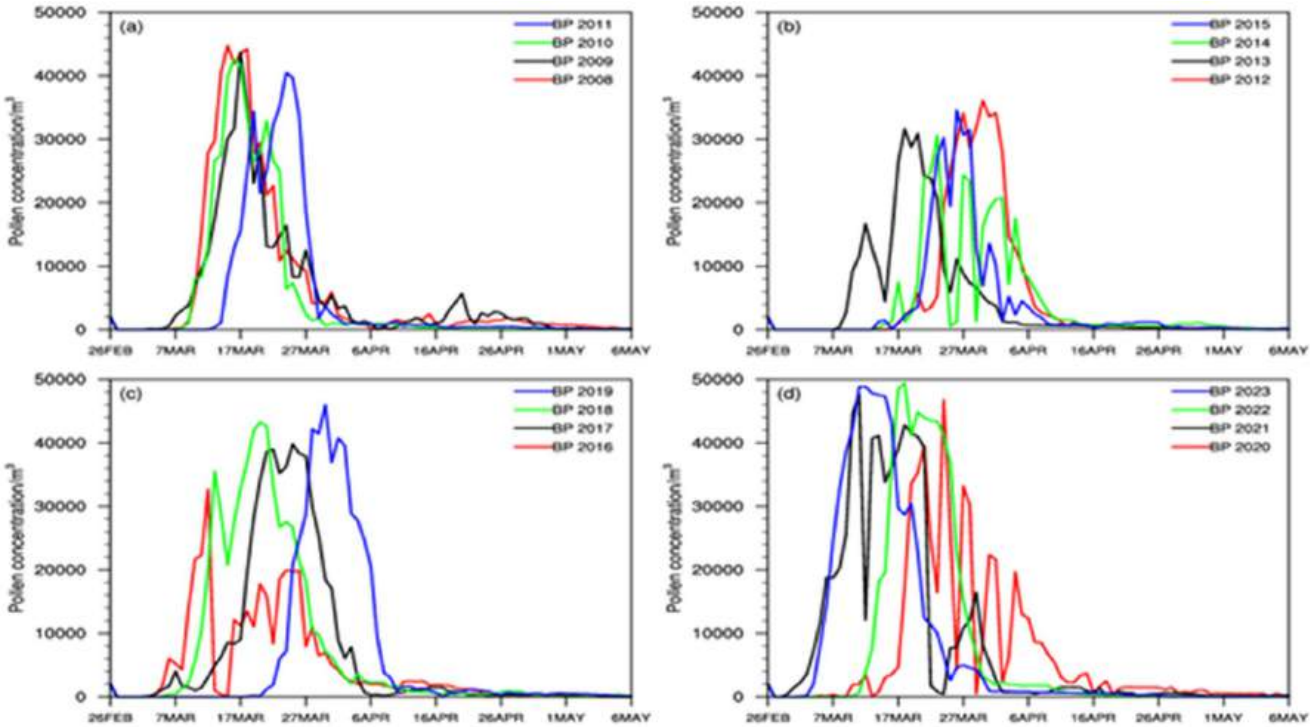

Fig. 7 B. *papyrifera* pollen concentrations in Islamabad during 2008–2023. The Y-axis presents pollen concentrations and the X-axis shows the pollen season time scale. BP codes for *Broussonetia* pollen. **a**, **b**, **c**, and **d** present 4-year data marked in different colors

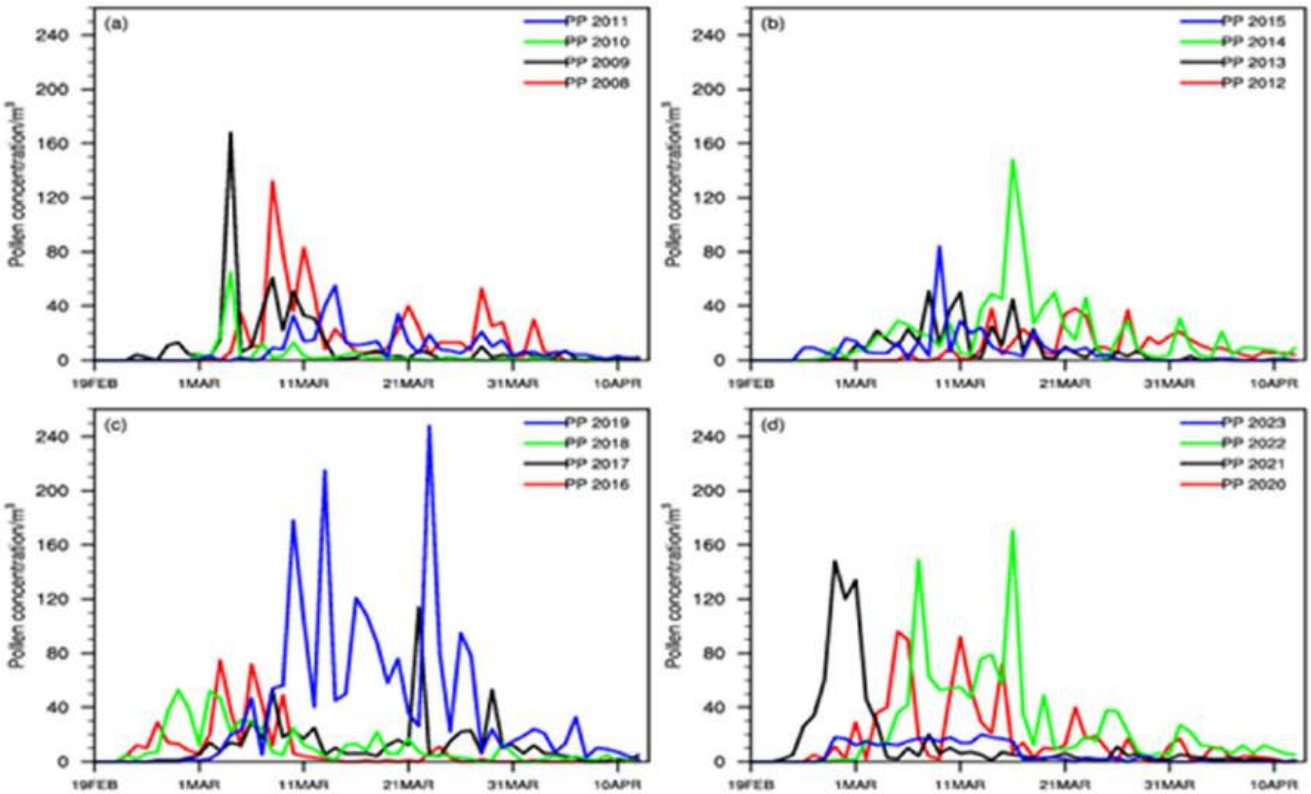

Fig. 8 *Pinus* pollen concentrations in Islamabad during 2008–2023. The Y-axis presents pollen/m^3 and the X-axis shows the pollen season time scale. PP codes for *Pinus* pollen. **a**, **b**, **c**, and **d** present 4 years of data marked in different colors

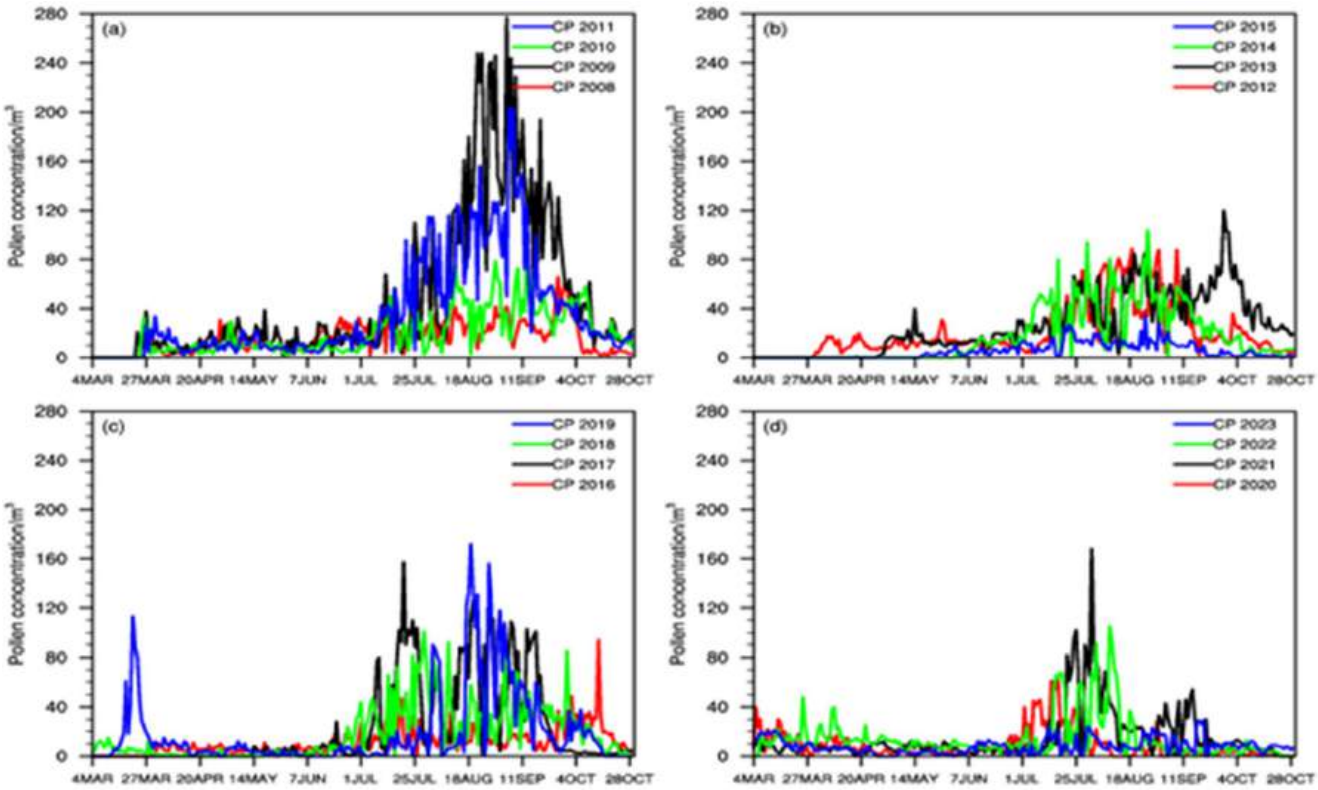

Fig. 9 *C. sativa* pollen concentrations in Islamabad during 2008–2023. The Y-axis presents pollen/m^3 and the X-axis shows the pollen season time scale. CP codes for *C. sativa* pollen. **a**, **b**, **c**, and **d** present 4-year data marked in different colors

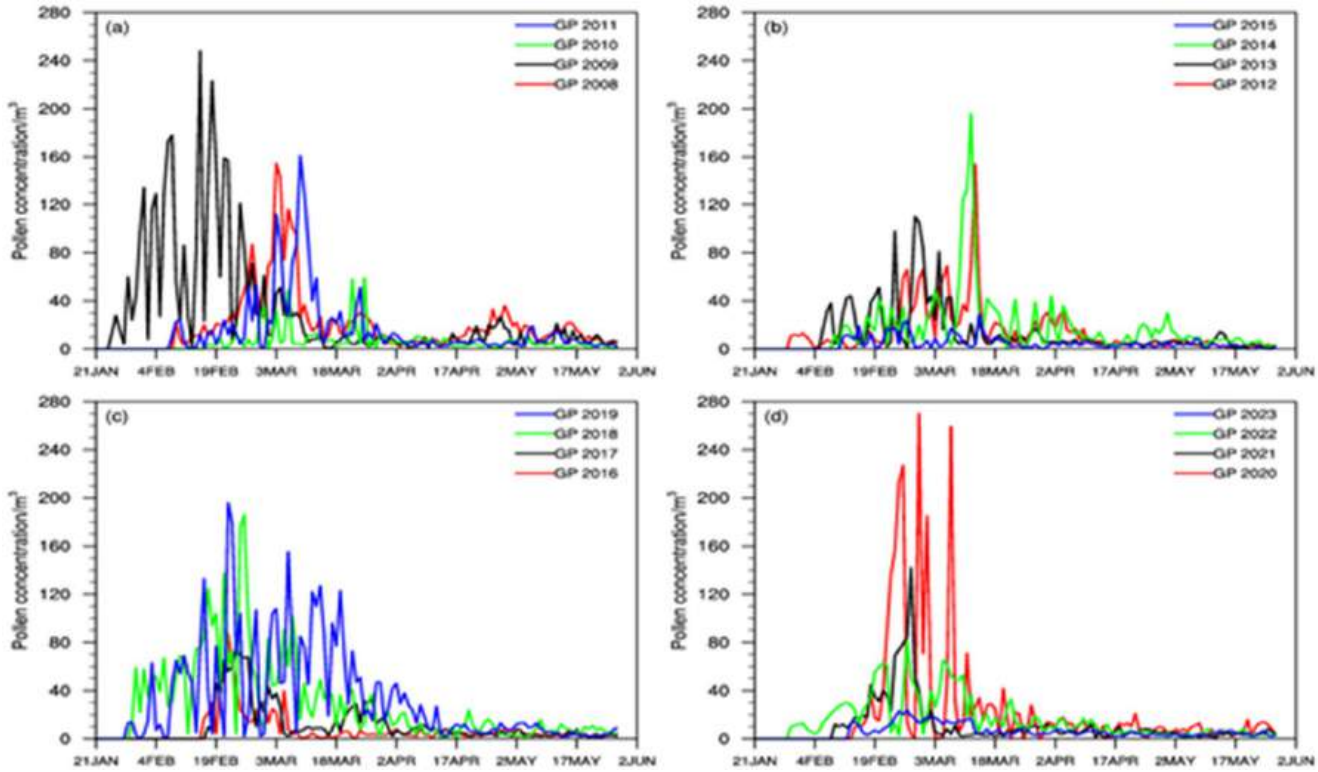

Fig. 10 Poaceae pollen concentrations in Islamabad during 2008–2023. The Y-axis presents pollen/m^3 and the X-axis shows the pollen season time scale. GP codes for Poaceae pollen. **a**, **b**, **c**, and **d** present 4-year data marked in different colors

driven by climate change, have also prolonged the pollen season and increased the prevalence of airborne allergens. For example, *P. pratense* produced 50% more pollen under elevated CO_2 conditions, independent of ozone (O_3) levels (Albertine et al. 2014), which can increase pollen allergies. Physicians treating airway allergies report an increase in allergies symptoms attributed to climate change (Sarfaty et al. 2015). Moreover, pollen exposed to low levels of CO, O_3, and SO_2 has been shown to enhance allergic responses. Immunodetection assays revealed increased IgE binding in sensitized individuals, indicating that air pollutants may intensify allergenicity (Cuinica et al. 2015).

Statistical correlation between pollen concentration and meteorological parameters

The results of this study indicate that *B. papyrifera*, *Pinus*, and Poaceae (grass) pollen concentrations are positively correlated with temperature, whereas *C. sativa* shows no such correlation. Poaceae and *Pinus* exhibited a weak negative correlation with relative humidity, while *C. sativa* demonstrated a weak positive correlation under humid conditions. Wind speed in the study area ranged from 0 to 3 m/s, predominantly blowing from the southeast, which likely facilitated the dispersion of both pollen and air pollutants within the region (Fig. 5). The mean monthly relative humidity in Islamabad was recorded at 60.4% in 2022 and 64.7% in 2023 (Fig. 2g). These findings align with previous studies, which report that relative humidity at or below 60% enhances the ability of pollen grains to adsorb atmospheric pollutants (Galveias 2021). Pollen grains are hygroscopic and absorb moisture from the air. As humidity increases, the grains become heavier and settle more rapidly due to gravitational forces. Consequently, higher relative humidity is generally associated with a decrease in airborne pollen concentrations. In this study, increasing humidity levels were negatively correlated with *B. papyrifera*, *Pinus*, and Poaceae pollen concentrations, and positively correlated with an increase in *C. sativa* pollen levels (pollen/m³). *Poaceae*, *C. sativa*, and *B. papyrifera* exhibited a weak negative correlation with monthly total rainfall, whereas *Pinus* showed a weak positive correlation (Fig. 5). Rainfall is known to remove airborne pollen by washing it out of the atmosphere and off floral surfaces, thereby reducing pollen concentrations near ground level (Ravindra et al. 2021).

Ravindra et al. (2021) reported similar trends using Spearman correlation coefficients for both total and seasonal pollen concentrations of *Parthenium hysterophorus*, *Poaceae*, *Celtis occidentalis*, *Morus alba*, *C. sativa*, and *Eucalyptus* species. In their study, *P. hysterophorus*, *Poaceae*, and *C. sativa* showed a significant positive correlation with temperature and a negative correlation with relative humidity. Precipitation was negatively correlated with annual pollen concentrations of *Eucalyptus* and *Poaceae*, while relative humidity was inversely correlated with pollen levels of *P. hysterophorus*, *M. alba*, *Poaceae*, *C. sativa*, and *Eucalyptus*. Pollen production is generally temperature-dependent; however, extreme temperatures can inhibit pollination, particularly in spring-flowering species such as *P. pratense* and *B. papyrifera*. Adverse weather conditions, including excessive heat or intense precipitation, may impose physiological stress on plants, further disrupting pollination processes.

Pollen season shift

This study identifies notable shifts in the pollen seasons of *B. papyrifera*, *Pinus*, *C. sativa*, and Poaceae in Islamabad (see Figs. 7, 8, 9 and 10). These changes are evident in terms of the SPIn, onset, and duration of the pollen seasons. The pollen seasons of *B. papyrifera* and *Pinus* have notably extended (see Fig. 6), with SPIn analysis revealing an overall increase in total airborne pollen concentrations over the study period. These findings are consistent with previous studies linking the extension of birch tree (*Betula* spp.) pollen seasons to climate change. For instance, Zhang et al. and Lam et al. reported similar trends (Zhang et al. 2013; Lam et al. 2024) with Lam et al. using a 52-year pollen monitoring dataset to demonstrate a strong correlation between rising seasonal temperatures and shifts in birch pollen phenology.

Our study revealed noticeable changes in the pollen seasons of *C. sativa* and Poaceae (grasses) in Islamabad (see Figs. 9 and 10). SPIn (Seasonal Pollen Integral) analysis showed a decline in total airborne pollen concentrations for both taxa over the study period. This decrease may be linked to anthropogenic activities, particularly urban expansion, which has led to significant reductions in vegetated land cover. According to Kamran et al. (2023), built-up areas on previously bare land in Islamabad increased from 5.7% (52 km²) to 25.7% (233 km²), while bare land decreased from 42.2% (382 km²) to 18.1% (164 km²) (Kamran et al. 2023). These land use changes are likely contributors to the declining SPIn, as the reduction in natural vegetation limits pollen production.

Our findings are consistent with those of Adams-Groom et al., who reported a decline in Poaceae SPIn and a reduction in the number of high-pollen days across the United Kingdom (Adams-Groom et al. 2022). Similarly, Ravindra et al. reported significant positive correlations between *C. sativa* and Poaceae pollen concentrations and temperature in India, further highlighting the influence of meteorological variables on pollen dynamics (Ravindra et al. 2021). Temperature, in particular, is a critical driver of pollen season onset; our data support the well-documented pattern that increasing temperatures, up to a species-specific threshold,

are positively correlated with earlier pollen season initiation. These findings align with broader global literature documenting climate-driven shifts in pollen phenology. For example, earlier *Betula* pollination and extended *Ambrosia artemisiifolia* pollen seasons have been observed at latitudes above 44°N (Ziska 2020). Many pollen taxa across Europe exhibit an increase in yearly airborne pollen/m^3 in urban areas in comparison to semi-urban and rural areas. A large dataset comprising 23 plant species was monitored in 97 locations in Europe during 1977–2009, which exhibited a 59% increase and a 41% decrease in annual pollen index (API) (Ziello et al. 2012). In a separate 21-year study monitoring 25 plant taxa, significant increases were observed in *A. artemisiifolia*, *Artemisia*, *Alnus*, and *Taxus*, while *Ulmus* pollen levels declined. The average onset of the pollen season advanced by approximately 0.84 days per year, with tree species exhibiting stronger responses to climate change than herbaceous (Clot 2003). Experimental studies have also confirmed that elevated CO_2 levels can increase both airborne pollen concentrations and allergenic potency (Ziello et al. 2012). Future projections under a Bayesian framework suggest that, between 2020 and 2100, pollen concentrations will rise significantly, with pollen season onset and peak dates advancing by approximately two weeks (Zhang et al. 2013). These changes have direct implications for human health, particularly in urban areas.

Islamabad, a rapidly urbanizing city, faces growing environmental challenges. Its topography, characterized by lower elevations in the southern and western sectors, contributes to the retention of both pollen and air pollutants in these regions. Poor air quality has well-documented health impacts with an increase in respiratory diseases linked to polluted air (Tang et al. 2022), while De Weger et al. found that high levels of allergenic pollen exacerbate symptoms in sensitized individuals (De Weger et al. 2021). The WHO 2021 AQG emphasizes the need for clean air to protect public health; however, air quality across South Asia continues to deteriorate (Nawaz 2022; Sharma and Mauzerall 2022; Torabi et al. 2023; Khan et al. 2023a). Recent assessments report elevated levels of particulate matter (PM_{10}, $PM_{2.5}$), sulfur dioxide (SO_2), nitrogen dioxide (NO_2), and ozone (O_3) in northern India (Sharma and Mauzerall 2022) and similarly poor air quality conditions in Pakistan, Afghanistan, and Iran (Taghizadeh et al. 2023; Torabi et al. 2023). Contributing factors include high population density, rapid urban development, and limited regulatory enforcement. Establishing comprehensive pollen and air quality monitoring systems across South Asia is essential for understanding the interaction between airborne pollen concentrations and air pollutants. Such efforts would support more accurate forecasting and mitigation of pollen allergy-related health risks in increasingly urbanized and polluted environments.

Conclusion

This study presents comprehensive evidence that the atmosphere of Islamabad contains air pollutant concentrations exceeding the limits recommended by the WHO and EPA. Notably, nitrogen dioxide (NO_2) and ozone (O_3) were positively correlated with increased airborne pollen concentrations across all studied species, while carbon dioxide (CO_2) showed a species-specific positive correlation with *B. papyrifera* pollen levels. Rainfall was consistently associated with reduced airborne pollen concentrations, while the effects of relative humidity varied among taxa. A key finding of this research is the statistically significant correlation between temperature and pollen season trends over the period 2008-2023. During this time, the SPIn of *B. papyrifera* and *Pinus* increased, whereas *C. sativa* and Poaceae exhibited a decline in SPIn. Additionally, shifts in pollen season onset, duration, and end dates were observed, indicating pollen phenology changes in response to climatic and environmental factors. Importantly, the increase in SPIn and the extension of the pollen season appear to be influenced by both meteorological parameters and air pollutant concentrations. These trends suggest that future changes in climate and air quality may further elevate airborne pollen levels and prolong the pollen season, thereby increasing population exposure to allergenic pollen and associated health risks. Future research should focus on quantifying the allergenic potential of pollen under elevated pollutant levels and assessing the combined effects on respiratory health. Regional collaboration in pollen monitoring and air quality management across South Asia is essential to mitigate the growing burden of allergenic pollen exposure and to promote a healthier atmosphere.

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Author contributions M.H. collected and analyzed the whole data, and wrote the original manuscript; K.U. provided air pollutants data and helped in plotting figures in NCL, and improved the manuscript; S.N. reviewed the strategy and improved the manuscript, S.B. provided pollen and meteorological datasets, N.A. provided necessary support from the National Institute of Health and Z.A. designed the strategy, reviewed the manuscript, and supervised the whole project.

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Data and code availability All data and NCL codes used in this study are available upon request from the corresponding author.

Declarations

Competing interests The authors declare no competing interests.

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